PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation

In partial fulfillment of the requirements for Capstone 1 and Research

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Table of Contents

CAPSTONE PROJECT PROPOSAL....1

Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation....1

ABSTRACT....6

CHAPTER I....7

INTRODUCTION....7

1.1 Background Of The Study....7

1.2 Statement of the Problem....8

General Problem....9

Specific Problems....9

1.3 Objectives of the Study....9

1.3.1 General Objectives....10

1.3.2 Specific Objectives....10

1.4 Scope and Delimitation....10

Delimitation of the Study....11

1.5 Significance of the Study....12

1.6 Definition of Terms....13

CHAPTER II....15

REVIEW OF RELATED LITERATURE AND STUDIES....15

2.1 Foreign Studies....15

2.2 Local Studies....23

2.3 Related Literature....31

2.4 Synthesis of the Reviewed Literature....39

REFERENCES....41

CHAPTER III....44

METHODOLOGY....44

3.1 Research Design....44

3.2 System Development Methodology....45

3.2.1 Planning Phase....45

3.2.2 Analysis Phase....46

3.2.3 Design Phase....47

3.2.4 Development and Implementation Phase 48
3.2.5 Testing Phase 49
3.2.6 Deployment and Maintenance Phase 50
3.3 System Architecture 51
3.3.1 Overview of the System 51
3.3.2 Client-Server Setup 52
3.3.3 User Interaction Flow 53
3.4 Tools and Technologies Used 54
3.4.1 Programming Language 54
3.4.2 Front-End Library 55
3.4.3 Styling Framework 55
3.4.4 Back-End Technologies 56
3.4.5 Database Management System 57
3.4.6 Email Notification Service 57
3.4.7 Code Editor and Development Tools 58
3.4.8 Version Control System 58
3.4.9 Summary of Tools and Technologies 59
3.5 Data Gathering Procedures 59
3.5.1 Interview Method 60
3.5.2 Observation Method 60
3.5.3 Document Analysis 61
3.5.4 Requirements Gathering and Validation 62
3.5.5 Importance of Data Gathering in the Study 63
3.6 Testing Procedures 64
3.6.1 Unit Testing 64
3.6.2 System Testing 66
3.6.3 User Acceptance Testing (UAT) 68
3.6.4 Bug Identification and Resolution 69
3.6.5 Summary of Testing Procedures 69
3.7 Evaluation Criteria 70
3.7.1 Functionality 70
3.7.2 Usability 70
3.7.3 Reliability 71

3.7.4 Efficiency **71**

3.7.5 Security **72**

3.7.6 Summary of Evaluation Criteria **73**

Data Flow Diagram (DFD) Level 0 ..... **73**

Entity Relationship Diagram (ERD) **75**

Use Case Diagram **79**

CHAPTER IV 82

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION 82

4.1 Introduction **82**

4.2 Overview of the Developed System **83**

4.3 System Development Methodology **85**

4.3.1 Requirements Gathering **85**

4.3.2 System Planning **85**

4.3.3 System Design **86**

4.4 System Architecture **91**

4.5 Hardware and Software Requirements **94**

4.5.1 Hardware Requirements **95**

4.5.2 Software Requirements 96

4.6 Database Design **97**

CHAPTER V 101

CONCLUSION AND RECOMMENDATIONS 101

5.1 Introduction **101**

5.2 Summary of Findings **101**

5.2.1 Digitalization of Manual Lot Monitoring **101**

5.2.2 Centralized Data Management **102**

5.2.3 Interactive Lot Availability Monitoring **102**

5.2.4 User Authentication and Access Control **102**

5.2.5 Customer and Transaction Management **102**

5.2.6 Dashboard and Reporting **103**

5.2.7 Email Notification **103**

5.2.8 Overall System Development **103**

5.3 Conclusion **103**

5.4 Recommendations **104**

5.4.1 System Improvements **104**

5.4.2 Future Research Directions **105**

5.5 Impact of the Study **106**

5.5.1 Organizational Impact **106**

5.5.2 Employee Impact **106**

5.5.3 Data Management Impact **106**

5.5.4 Technological Impact **107**

5.5.5 Academic Impact **107**

5.6 Summary **107**

ABSTRACT

The accurate monitoring and management of real estate lot information are essential for maintaining organized property records and determining lot availability. Golden Dragon Estate Corporation previously relied on printed subdivision maps and manual color coding to monitor lot statuses, which made updating and tracking property information less convenient. This study focused on the design and development of a Web-Based Lot Tracking System for Golden Dragon Estate Corporation to improve the monitoring and management of lot information.

The developed system provides a centralized platform for authorized employees to view and manage lot records through an interactive digital subdivision layout. It uses color-coded indicators to identify lot statuses, with green representing available lots, yellow representing pending lots, and red representing sold lots. The system also supports lot information management, status updates, and centralized recordkeeping. React and Tailwind CSS were used to develop the user interface, while Node.js and Express.js handled server-side operations. MySQL was used to store and manage the system's data. The Agile methodology guided the system development process, including requirements gathering, system design, implementation, and testing.

The system provides a digital alternative to manually updating printed subdivision maps and helps authorized employees access organized lot information. It supports more efficient lot status monitoring and helps reduce the difficulties associated with maintaining manual records. The developed system serves as a tool for improving property information management and supporting daily lot monitoring operations at Golden Dragon Estate Corporation.

Keywords: Digital Lot Tracking, Real Estate Management, Lot Status Monitoring, Web-Based System, Interactive Map

CHAPTER I

INTRODUCTION

1.1 Background Of The Study

The real estate industry plays a significant role in economic development by providing residential, commercial, and industrial properties to individuals and organizations. As the demand for real estate continues to grow, companies are required to manage property information efficiently to ensure accurate transactions and effective customer service. One important aspect of real estate management is monitoring the availability and status of residential lots. Accurate information regarding lot availability enables companies to respond promptly to customer inquiries, prevent double-selling, and maintain organized records of property transactions.

Golden Dragon Estate Corporation is a real estate company operating in Iloilo Province, offering residential lots in various locations such as Oton, Guimbal, and neighboring municipalities. The company currently relies on traditional methods for monitoring lot availability. Printed subdivision layout maps are used to represent residential lots, and employees manually indicate lot status through color-coding. Green represents available lots, yellow represents pending reservations, and red represents sold lots.

Although this method has been used for several years, it presents several operational challenges. Updating lot statuses requires employees to manually modify physical documents, which can be time-consuming and inefficient. As the number of lots and subdivision projects increases, maintaining accurate and updated records becomes more difficult. Furthermore, physical documents are vulnerable to damage, loss, misplacement, and deterioration over time. These issues may result in inaccurate information, delays in customer service, and potential business inefficiencies.

Advancements in information technology have enabled organizations to automate manual processes through web-based information systems. Such systems improve efficiency, accuracy, accessibility, and data security while reducing dependence on paper-based records. By digitizing lot monitoring operations, real estate companies can maintain real-time information regarding property availability and streamline internal processes.

To address the challenges encountered by Golden Dragon Estate Corporation, this study proposes the development of a Web-Based Lot Availability Monitoring System. The system will allow authorized employees to securely manage and update lot statuses through an interactive digital platform. Instead of manually coloring printed layout maps, users can simply select a lot and update its status using the system interface. The updated information will then be reflected immediately within the digital subdivision map.

The proposed system will be developed using React, Tailwind CSS, Node.js, Express.js, and MySQL. Additionally, Nodemailer SMTP integration will be utilized for email notification functionalities. Through the implementation of this system, Golden Dragon Estate Corporation can improve operational efficiency, reduce the risks associated with physical records, and enhance the accuracy of lot monitoring activities.

1.2 Statement of the Problem

Golden Dragon Estate Corporation is engaged in the sale and management of residential lots across different locations within Iloilo Province, including municipalities such as Oton and Guimbal. As part of its daily operations, the company continuously monitors the availability and status of residential lots to ensure accurate information is provided to employees, management, and prospective buyers. Maintaining updated records of available, pending, and sold lots is essential for effective decision-making and efficient business operations. At present, the company utilizes a traditional and manual method of monitoring lot availability. Printed subdivision layout maps are used as the primary reference for tracking lot statuses.

Employees manually indicate the status of each lot by applying color codes on the physical maps. Green is used to represent available lots, yellow indicates pending transactions or reservations, and red signifies sold lots. While this method has been functional for many years, it presents several limitations that affect the company's operational efficiency.

One of the major concerns associated with the current system is the time and effort required to update records. Every transaction involving a lot requires employees to manually revise the printed maps, increasing the possibility of delays and inconsistencies in information. Furthermore, physical documents are vulnerable to loss, damage, deterioration, and unauthorized alterations. As the company continues to manage multiple subdivision projects and an increasing number of lot records, the challenges associated with manual monitoring become more evident.

The absence of a centralized digital system also limits the accessibility and reliability of information. Employees must rely on physical records when verifying lot availability, which can make the retrieval of information slower and less efficient. Additionally, there is a greater possibility of human error in recording, updating, and maintaining lot information. These issues may negatively affect customer service, internal operations, and overall productivity.

In response to these challenges, the researchers propose the development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. The proposed system aims to automate the monitoring and management of lot information through a secure and centralized platform. By replacing the manual process with a digital solution, the company can improve efficiency, enhance data accuracy, strengthen security, and facilitate easier access to information.

General Problem

How can a Web-Based Lot Availability Monitoring System be developed and implemented to improve the efficiency, accuracy, security, accessibility, and reliability of monitoring residential lot availability at Golden Dragon Estate Corporation?

Specific Problems

Specifically, this study seeks to answer the following questions:

1. The current lot monitoring process relies heavily on printed subdivision maps and manual color-coding methods.
2. Updating lot statuses requires considerable time and effort from employees, resulting in operational inefficiencies.
3. Physical records are susceptible to loss, damage, deterioration, and misplacement.
4. The existing process increases the likelihood of human errors during the recording and updating of lot information.
5. The company lacks a centralized database for organizing and managing lot records.
6. Access to lot information is limited to physical documents, making information retrieval less convenient and efficient.
7. Monitoring lot availability across multiple subdivision projects becomes increasingly difficult as records grow.
8. The current process does not provide real-time updates regarding changes in lot status.
9. There is no secure authentication mechanism that restricts access to authorized personnel only.
10. The company lacks an automated system capable of maintaining accurate and organized lot information.
11. The existing method makes it difficult to maintain consistency and reliability in monitoring available, pending, and sold lots.
12. The current process does not utilize modern information technology solutions that could improve operational efficiency.

1.3 Objectives of the Study

The primary purpose of this study is to develop a modern technological solution that addresses the challenges encountered by Golden Dragon Estate Corporation in monitoring residential lot availability. Through the implementation of a web-based information system, the researchers aim to improve the company's operational processes by automating manual tasks and providing a more efficient method of managing lot records.

The proposed system seeks to transform traditional paper-based monitoring procedures into a centralized digital platform capable of providing accurate, reliable, and real-time information regarding lot availability. By integrating database management, authentication mechanisms, and interactive monitoring features, the system is expected to improve overall organizational productivity and data management practices.

1.3.1 General Objectives

The general objective of this study is to design, develop, and implement a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation that will automate the monitoring, management, and updating of residential lot information while improving operational efficiency, data accuracy, accessibility, and security.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Design and develop a web-based lot availability monitoring system for Golden Dragon Estate Corporation.
2. Create a centralized database for storing, managing, and retrieving lot information efficiently.
3. Develop an interactive digital subdivision layout that visually represents lot availability.
4. Automate the process of updating lot statuses from available to pending or sold.
5. Implement secure user authentication and authorization features for authorized employees.
6. Allow users to efficiently search, filter, and retrieve lot information.
7. Provide real-time updates whenever changes are made to lot records.
8. Reduce the organization's dependence on paper-based records and manual monitoring procedures.
9. Improve the accuracy and consistency of lot information management.

1.4 Scope and Delimitation

This study focuses on the development of a Web-Based Lot Availability Monitoring System specifically designed for Golden Dragon Estate Corporation. The proposed system will serve as a centralized platform for monitoring, managing, and updating residential lot information across the company's subdivision projects.

The system will provide functionalities that support the daily monitoring activities of authorized employees and administrators. Through the use of modern web technologies such as React, Tailwind CSS, Node.js, Express.js, and MySQL, the system will provide a secure and efficient environment for managing lot records.

Specifically, the system will include the following features:

- User login and authentication
- Role-based access control
- User account management
- Lot information management
- Interactive subdivision layout visualization
- Lot status monitoring
- Status updates for available, pending, and sold lots
- Search and filtering capabilities
- Database storage and management
- Email notification integration through Nodemailer SMTP
- Administrative monitoring functions
- Secure record management

The proposed system will be accessible through web browsers and will be deployed as a web-based application

Delimitation of the Study

Despite its intended benefits, the proposed system has several limitations.

1. The system focuses exclusively on monitoring residential lot availability.
2. The system is intended only for internal organizational use.
3. Access will be limited to authorized employees, administrators, and management personnel.
4. Clients and prospective buyers will not have direct access to the system.
5. The system will not support online reservations or online purchasing of lots.
6. Payment processing and financial transactions are excluded from the system.
7. Accounting and bookkeeping functionalities are beyond the scope of the study.
8. Legal document management and contract processing are not included.
9. Mobile application development is not part of the project.
10. GIS-based mapping and satellite visualization technologies will not be implemented.
11. The system will only cover subdivision projects managed by Golden Dragon Estate Corporation.
12. Predictive analytics, artificial intelligence, and sales forecasting functionalities are not included.

Target Users

The intended users of the proposed system include:

- System Administrators
- Authorized Employees
- Management Personnel

Platform

The proposed system will be developed as a web-based application accessible through desktop and laptop web browsers connected to the internet or a local network.

1.5 Significance of the Study

The successful development and implementation of the proposed Web-Based Lot Availability Monitoring System is expected to provide significant benefits to Golden Dragon Estate Corporation and its stakeholders. The system aims to improve operational efficiency, enhance data accuracy, and reduce the challenges associated with manual record management.

Golden Dragon Estate Corporation

The organization will benefit from a more efficient and reliable method of monitoring residential lot availability. The proposed system will improve record management, reduce operational delays, and support the company's efforts to modernize its business processes.

Administrators

Administrators will benefit from having centralized control over system operations, user accounts, and lot records. The system will enable them to efficiently manage information while maintaining security and accuracy.

Employees

Authorized employees will experience a more convenient method of updating and monitoring lot statuses. The automation of manual processes will reduce workload, save time, and improve productivity.

Management Personnel

Management personnel will gain access to organized and updated information regarding lot availability. This will support decision-making, monitoring activities, and strategic planning efforts.

Clients and Prospective Buyers

Although clients will not directly access the system, they will indirectly benefit from faster and more accurate information regarding lot availability, resulting in improved customer service.

Researchers

The study will provide researchers with valuable knowledge and practical experience in software development, web technologies, database management, systems analysis, and project implementation.

Future Researchers

Future researchers may utilize this study as a reference for developing similar systems related to property management, monitoring systems, and web-based information systems.

Future Developers

Future developers may use the system design, methodologies, and findings of this study as a foundation for creating enhanced versions of the system with additional features and capabilities.

1.6 Definition of Terms

Administrator

A user who possesses the highest level of access privileges and is responsible for managing system operations, user accounts, and records.

Authentication

The process of verifying the identity of a user before granting access to the system.

Authorization

The process of determining the access rights and permissions granted to a user after successful authentication.

Centralized Database

A database that stores information in a single location where data can be managed, accessed, and updated efficiently.

Database

An organized collection of data stored electronically for easy access, retrieval, and management.

Express js

A web application framework for Node.js used to build server-side functionalities and application programming interfaces (APIs).

Lot Availability

The current status of a residential lot indicating whether it is available, pending, or sold.

Monitoring System

A computerized application designed to track, manage, and display information for monitoring purposes.

Node js

A JavaScript runtime environment used to develop scalable server-side applications.

Nodemailer SMTP

A Node.js module that enables applications to send emails through the Simple Mail Transfer Protocol (SMTP).

React

A JavaScript library used for building interactive and dynamic user interfaces.

Residential Lot

A parcel of land intended for residential development and offered for sale by a real estate company.

Tailwind CSS

A utility-first CSS framework used for designing responsive and visually appealing web interfaces.

User Authentication

A security mechanism that validates user credentials before allowing access to the system.

Web-Based Application

A software application that operates through a web browser and can be accessed via a network connection.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The advancement of information technology has significantly influenced the real estate industry worldwide. Researchers have developed various web-based and database-driven systems to improve property management, information accessibility, and operational efficiency. These studies provide valuable insights into the effectiveness of digital technologies in managing real estate information and serve as a foundation for the development of the proposed Web-Based Lot Availability Monitoring System.

Development and Implementation of a Real Estate Web Application

The real estate industry has experienced significant technological advancements in recent years, driven by the growing demand for efficient property management and digital accessibility. To address the challenges associated with traditional property management practices, **Kumar [1]** developed and implemented a Real Estate Management System (REMS) that utilizes a web-based platform to manage property-related information. The study emphasized the importance of integrating digital technologies into real estate operations to improve information management and reduce inefficiencies associated with conventional recordkeeping and property monitoring methods.

According to **Kumar [1],** traditional real estate management practices often involve manual documentation, fragmented information storage, and localized recordkeeping systems. These practices may result in delayed information updates, data inconsistencies, communication gaps, and difficulties in maintaining accurate property records. To address these challenges, the study introduced a web application that centralizes property information within a unified database. This approach enables users to access, update, and manage property records more efficiently, supporting improved information accessibility and decision-making.

The developed Real Estate Management System incorporated functionalities such as property listing management, property status updates, user account management, and information retrieval. These features were intended to streamline administrative processes by reducing manual intervention and providing users with access to property information through a centralized platform. The study also highlighted the importance of maintaining updated property descriptions, pricing information, availability statuses, and other essential records to support effective property management.

A significant contribution of the study was its emphasis on the use of centralized databases and web-based technologies to improve the accessibility and consistency of real estate information. The system demonstrated the

potential of digital platforms to reduce administrative workload, minimize errors associated with manual data handling, and improve information management. Furthermore, the study highlighted how web applications can support organizational operations through more organized and accessible property records.

The work of **Kumar [1]** is relevant to the present study because both projects utilize web-based technologies to improve real estate information management. The Web-Based Lot Tracking System for Golden Dragon Estate Corporation addresses the challenges associated with manually maintained and color-coded subdivision maps by providing a digital platform for monitoring residential lot availability. Similar to the system discussed by Kumar, the present system centralizes property information and allows authorized personnel to manage lot records electronically.

However, while Kumar's study focuses on broader real estate management functions, the present study specifically emphasizes lot status monitoring and the visualization of lot availability through an interactive digital subdivision layout. This distinction establishes the specific focus of the current research while demonstrating its relationship to existing approaches to web-based property management. The related study provides a conceptual basis for examining how centralized information management and web technologies can support more organized and efficient lot monitoring operations at Golden Dragon Estate Corporation.

Estate-Manager: A JWT-Secured Real Estate Platform with Role-Based Access Control

The security and integrity of property-related information are important considerations in the digital transformation of the real estate industry. The adoption of web-based information systems introduces security concerns involving unauthorized access, data manipulation, and the protection of sensitive records. To address these concerns, **Kushal B. C. [2]** developed Estate Manager, a web-based real estate platform that integrates property management functionalities with security mechanisms designed to protect information and regulate user access.

The study highlighted the limitations of fragmented workflows and manual recordkeeping, which may contribute to inconsistent data management, limited accountability, and human errors. The absence of appropriate access control mechanisms may also expose property records to unauthorized modifications. To address these challenges, Estate-Manager adopted a centralized platform intended to improve property information management while strengthening data security.

Estate-Manager was developed using a full-stack architecture and a MySQL database management system. Its security features include JSON Web Token (JWT) authentication and Role-Based Access Control (RBAC). JWT authentication verifies user identity and supports authenticated access to protected application resources. Meanwhile, RBAC regulates access to system functionalities according to users' assigned roles and permissions. These mechanisms help restrict sensitive operations, including creating, editing, updating, and deleting property records, to users with appropriate authorization.

The study also examined system performance, particularly data retrieval and filtering. According to the provided study description, Estate Manager achieved an average query response time of less than 100 milliseconds. This reported result highlights the potential of integrating security mechanisms with efficient database operations in web-based property management applications. However, the result should be interpreted within the context of the system's reported testing conditions.

The work of **Kushal B. C. [2]** is relevant to the present study, the Web-Based Lot Availability System for Golden Dragon Estate Corporation, because both systems emphasize centralized property information management and controlled access to real estate records. In the present system, accurate lot status updates are essential for distinguishing available, pending, and sold properties. Appropriate authentication and authorization mechanisms can help ensure that only authorized personnel perform sensitive operations and that property information remains protected from unauthorized changes.

Although Estate-Manager addresses broader property management functions, the present study focuses specifically on lot availability monitoring through an interactive digital subdivision layout. Nevertheless, both studies share the objectives of improving information management, maintaining data integrity, and supporting operational efficiency through web-based technologies. Estate-Manager therefore provides a relevant basis for examining the role of authentication, role-based permissions, and centralized database management in the development of a reliable lot monitoring system.

Innovative Real Estate Management System: Artificial Intelligence-Based Segregation

The advancement of information technology has contributed to the development of web-based applications that improve property information management, data accessibility, and administrative decision-making in the real estate industry. **Joshi et al. [3]** presented an Innovative Real Estate Management System that incorporates modern web technologies and artificial intelligence-based techniques to support the organization and management of property information. The study addressed the challenges associated with managing increasing volumes of property records and emphasized the importance of developing responsive and scalable applications for real estate operations.

According to the study, traditional property management approaches may become inefficient as the volume of property records, user interactions, and transactions increases. To address these challenges, the researchers employed a JavaScript-based technology stack consisting of React, Node.js, and Express.js. These technologies support the development of interactive user interfaces and server-side applications capable of processing requests and managing property-related information. The use of asynchronous, event-driven operations in Node.js also supports applications that require frequent data updates and continuous user interaction.

The system incorporated user authentication mechanisms to regulate access to property information and protect sensitive records. It also featured interactive dashboards and data visualization tools that presented property-related information through graphical representations and summarized reports. These features were intended to

help administrators interpret operational data, monitor property information, and support decision-making. The study highlighted the potential of combining modern web technologies, security mechanisms, and visual reporting tools to improve the management of real estate information.

The study by **Joshi et al. [3]** is relevant to the present research, the Web-Based Lot Availability System for Golden Dragon Estate Corporation, because both systems apply web technologies to improve the accessibility and management of property records. The use of interactive dashboards in the related study provides a useful reference for presenting summarized lot information and enabling authorized personnel to monitor property statuses efficiently. In the present system, the digital subdivision layout allows users to identify available, pending, and sold lots through visual indicators.

Furthermore, the study's discussion of authentication and responsive web architecture highlights important considerations for developing a system that manages frequently updated property records. Although the related study addresses broader real estate management functions and artificial intelligence-based segregation, the present research focuses specifically on lot availability monitoring and status management. This difference establishes the scope of the current project while demonstrating its relationship to existing web-based real estate solutions. Therefore, the study provides a relevant reference for understanding the potential contributions of modern web technologies, interactive dashboards, and access control mechanisms to property information management.

Property Rental and Management System

The increasing demand for efficient property management solutions has encouraged real estate organizations to adopt digital technologies that improve operational efficiency and information accessibility. In a study by **Salunkhe [4]**, a Property Rental and Management System was developed to modernize property administration through web technologies and centralized database management. The study addressed the limitations of traditional paper-based processes, particularly in storing, retrieving, and maintaining property-related information.

According to Salunkhe [4], conventional property management practices often involve manual documentation and fragmented recordkeeping, which may result in delayed information retrieval, duplicated records, increased administrative workload, and human errors. Physical documents may also be vulnerable to loss, damage, and unauthorized access. To address these challenges, the study introduced a centralized web-based platform for storing, managing, and retrieving property information. This approach supports more organized recordkeeping and improves access to information.

The developed system utilized JavaScript-based technologies and a MySQL relational database management system. Node.js supported server-side request processing, while the centralized database provided a common repository for property records. This architecture enabled users to access and manage information through a web-based application. By maintaining records in a centralized environment, the system aimed to reduce inconsistencies associated with multiple copies of documents and simplify information maintenance.

The study emphasized the benefits of centralized data management, including improved information accessibility, more organized records, and reduced dependence on manual administrative procedures. It also highlighted the importance of maintaining updated property information so that authorized users could access current records when performing management tasks. These capabilities demonstrate how web-based applications can support property administration by providing a more accessible and organized alternative to conventional recordkeeping practices.

The work of Salunkhe [4] is relevant to the present study, *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*, because both studies address the limitations of manual property information management through centralized digital platforms. The related study provides a reference for understanding how web technologies and database management can support the storage, retrieval, and updating of real estate records.

Furthermore, the use of Node.js and MySQL described in the related study provides a useful technological reference for the development of database-driven property applications. In the present study, centralized lot records and digital status updates support the monitoring of available, pending, and sold residential lots. These functions address the challenges associated with manually maintained and color-coded subdivision maps by allowing authorized personnel to access and manage lot information through a digital platform.

Although Salunkhe [4] focused primarily on property rental and management operations, the present study concentrates specifically on residential lot availability monitoring and status management. Despite their differences in scope, both studies emphasize centralized information storage, improved accessibility, and more efficient property record management. Therefore, Salunkhe's work provides a relevant basis for examining how web-based technologies and centralized databases can support the development of the proposed lot availability monitoring system.

PROPVISTA: Real Estate Website

The advancement of web technologies has transformed the way real estate organizations manage property information and interact with users. Modern real estate applications increasingly incorporate interactive interfaces, centralized information management, automated notifications, and reporting tools to improve operational efficiency. Vijayalakshmi et al. [5] developed PROPVISTA, a web-based real estate platform designed to support property exploration, information management, and administrative operations through contemporary web development technologies.

The study highlighted the limitations of traditional property management practices, particularly the need for manual monitoring and administrative intervention to maintain accurate records and communicate important updates. These practices may contribute to delays, increased workloads, and errors in information dissemination. To address these challenges, PROPVISTA employed a full-stack architecture consisting of React for the front-end

interface, Express.js for server-side routing, and Node.js for backend processing. This architecture supported the development of an interactive web application capable of handling user requests and property information updates.

Another feature discussed in the study was the use of dashboards and data visualization tools to present summarized property information. These tools enabled administrators to review records through organized displays and graphical representations, supporting information monitoring and decision-making. The study also described the integration of user authentication mechanisms to restrict access to authorized users and help protect property-related information. In addition, automated notifications were incorporated to communicate important updates and reduce dependence on manual communication procedures.

The study emphasized the potential benefits of combining interactive interfaces, dashboard reporting, authentication, and automated notifications within a real estate management platform. These features can help organize information, improve communication, and support administrative activities. However, the extent of these benefits depends on the actual implementation and evaluation of the system.

The work of Vijayalakshmi et al. [5] is relevant to the present study, *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*, because both systems use web technologies to improve property information management. The use of dashboards in PROPVISTA provides a useful reference for presenting summarized information about available, pending, and sold lots. Through visual representations of lot statuses, authorized personnel can identify property availability more efficiently and access organized information for monitoring purposes.

Furthermore, the study's discussion of automated notifications is relevant to the notification functionality of the present system. The use of Nodemailer with SMTP, if implemented as described in the current system, supports the automated delivery of email notifications when specified lot-related activities occur. This functionality can reduce the need for manual communication and help keep designated recipients informed of important updates. The authentication mechanisms discussed in PROPVISTA also provide a relevant reference for protecting lot records and restricting access to authorized users.

Although PROPVISTA addresses broader real estate activities, including property exploration and management, the present study focuses specifically on residential lot availability monitoring and status management. Both studies nevertheless share the objectives of improving information accessibility, supporting administrative processes, and using web technologies to organize property records. Therefore, PROPVISTA provides a relevant reference for examining the application of interactive dashboards, automated notifications, and user authentication in the development of the lot availability monitoring system for Golden Dragon Estate Corporation.

Real Estate Website

The development of real estate management systems requires appropriate technologies and a structured software development methodology to address changing user requirements and operational needs. Wagh [6] presented a web-based Real Estate Management System developed using the Agile Software Development Methodology. The study focused on providing a centralized platform for managing property information while supporting data integrity and controlled user access.

The study identified several limitations of traditional property management practices, including manual record maintenance, delayed information updates, and difficulties in monitoring property inventory. To address these challenges, the system utilized a MySQL database to centralize property records and support the management of inventory information. Centralized data storage provides a structured approach to maintaining property records and allows authorized users to access updated information through the application.

A significant aspect of the study was the adoption of the Agile Software Development Methodology. Wagh [6] discussed the value of iterative development, continuous testing, and stakeholder feedback throughout the software development process. By organizing development activities into manageable iterations, developers can review system functionality and incorporate necessary improvements based on user requirements. This approach provides flexibility when refining system features and addressing operational needs during development.

The study also emphasized the importance of controlling access to property information. Role-Based Access Control (RBAC) was incorporated to assign permissions according to users' responsibilities. This approach restricts sensitive operations to users with appropriate authorization and helps protect property records against unauthorized changes. Such controls are particularly important in systems that allow multiple users to access and modify inventory information.

The work of Wagh [6] is relevant to the present study, *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*, because both projects address the management of real estate information through centralized digital platforms. The use of Agile provides a methodological reference for developing and refining system features in response to stakeholder requirements. In the present study, an iterative development approach can support improvements to lot monitoring, status updates, and other implemented functionalities.

Furthermore, the related study's use of MySQL provides a relevant reference for organizing and maintaining property records in a centralized database. This concept is applicable to the present system, which manages residential lot information and monitors available, pending, and sold statuses. Similarly, the discussion of role-based permissions highlights the importance of ensuring that only authorized personnel can perform restricted operations, helping maintain the integrity of lot records.

Although Wagh's study addresses broader real estate management functions, the present research specifically focuses on residential lot availability monitoring through a digital subdivision layout. Both studies share common concerns involving centralized information management, inventory monitoring, data consistency, and controlled

access. Therefore, Wagh's work provides a relevant methodological and technical reference for the development of the lot availability monitoring system for Golden Dragon Estate Corporation.

Web-Based Real Estate Management System: DBMS-Supported Solution for Property Listings and Buyer-Seller Interaction

The increasing complexity of real estate operations has created a demand for digital systems that efficiently manage property information while maintaining data accuracy and accessibility. Traditional property management practices often rely on manual records, paper-based documentation, and disconnected information repositories, which may result in inconsistent records, delayed information retrieval, and operational inefficiencies. To address these challenges, Paidi [7] developed a Web-Based Real Estate Management System supported by a Database Management System (DBMS) to facilitate property listing management and improve information handling.

The study emphasized the limitations of fragmented data storage, particularly the risks of duplicated records, inconsistent information, and delays in retrieving property details. To address these issues, the system employed a centralized relational database that organized property information within a unified environment. This approach provided a structured method for storing, maintaining, and retrieving records while supporting consistency across the application.

A notable feature of the system was its “Sale a Plot” module, which allowed authorized users to enter and manage property information through structured forms. The module incorporated data validation procedures to check the accuracy and completeness of submitted information before storing records in the database. Property details, including registration numbers, location information, dimensions, and pricing categories, were among the data elements identified in the study. These validation procedures were intended to reduce data entry errors and maintain the integrity of property records.

The system also adopted a three-tier architecture consisting of the presentation, application, and database layers. Separating these components helped organize user interface functions, application logic, and data management operations. The study reported query response times of less than 300 milliseconds during its performance testing, demonstrating the system's reported responsiveness under the evaluated conditions. However, this result should be interpreted within the context of the original study's testing environment.

The work of Paidi [7] is relevant to the present study, *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*, because both projects address the need for accurate, accessible, and organized real estate information. The related study's use of centralized database management provides a useful reference for maintaining lot records in a unified data repository. In the present system, this approach supports the management of lot numbers, block numbers, project information, and availability statuses.

Furthermore, the data validation procedures discussed by Paidi [7] are relevant to maintaining accurate lot information. Appropriate validation of lot identifiers and status updates can help prevent incomplete or invalid

entries from being stored in the system. The three-tier architecture described in the related study also provides a useful conceptual reference for separating the user interface, application processing, and database operations. Where implemented in the present system, React supports the presentation layer, Node.js and Express.js handle application operations, and MySQL manages persistent data storage.

Although Paidi's study focuses on property listings and buyer-seller interactions, the present research concentrates specifically on residential lot availability monitoring through a digital subdivision layout. Both studies share common objectives involving centralized data storage, information accuracy, and efficient record management. Therefore, Paidi's work provides a relevant reference for the database design, data validation, and architectural considerations associated with developing the lot availability monitoring system for Golden Dragon Estate Corporation.

2.2 Local Studies

Development and Evaluation of an Advanced Property Management System Using Unified OOAD for Optimizing Real Estate Operations

Encarnacion et al. [8] (2025) developed and evaluated an Advanced Property Management System (PMS) to improve real estate operations in Siargao Island, Philippines. The study addressed challenges associated with traditional property management practices, including delayed information retrieval, inefficient record handling, and difficulties in monitoring property transactions. To address these concerns, the researchers developed a web-based platform that centralized property information and provided users with more efficient access to property records.

The researchers applied the Unified Process for Object-Oriented Analysis and Design (OOAD) in developing the system. Its features included property directories, advanced search capabilities, status filtering, and a centralized database for storing, retrieving, updating, and managing property information. By integrating these functions into a single platform, the system provided a more organized approach to property record management and helped streamline real estate operations.

To evaluate the system’s quality, the researchers used the ISO/IEC 25010 Software Quality Standards. Information technology experts and prospective users assessed the system based on software quality characteristics, including functional suitability, security, usability, reliability, and performance efficiency. The study reported an overall mean rating of 4.52 out of 5.00, with ratings of 4.48 for functional suitability and 4.53 for security. These results indicated a high level of acceptance among the evaluators and suggested that the system addressed important user requirements while considering the protection of property information.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both systems use web-based technology and centralized databases to

improve the management of real estate information. However, their scopes differ. Encarnacion et al. focused on broader property management operations, including property directories and search functions, whereas the present study specifically focuses on monitoring residential lot availability and maintaining updated lot status information. The features of the local system provide a useful reference for organizing property records and making relevant information accessible to authorized users.

Furthermore, the study's use of ISO/IEC 25010 provides a reference for evaluating the quality of the present system, particularly in terms of functional suitability, security, usability, reliability, and performance efficiency. Its reported evaluation results also highlight the importance of assessing whether a system meets user requirements and protects organizational information. Overall, this local study provides a relevant basis for developing and evaluating a web-based solution tailored to Golden Dragon Estate Corporation's lot availability monitoring needs.

Web-Based Inventory Management System

Tanaman et al. [9] (2023) developed a Web-Based Inventory Management System to address operational challenges experienced by a commercial enterprise with multiple branches in Pagadian City, Zamboanga del Sur. The study focused on the limitations of manual, paper-based inventory tracking, which contributed to delayed reporting, inconsistent records, and difficulties monitoring inventory across different business locations. These challenges highlighted the need for a centralized system that could improve the accuracy, accessibility, and management of organizational information.

To address these concerns, the researchers designed and implemented a web-based platform that centralized inventory records and supported electronic data entry, retrieval, updating, and reporting. By replacing manual documentation with a digital database, the system provided a more organized approach to record management. The centralized platform also allowed authorized personnel to access inventory information, supporting coordination and helping management make decisions based on available records.

The study emphasized the role of automated data processing in reducing recording errors and duplicated entries. Centralizing information also helped maintain consistency across the enterprise's branches. According to the supplied study summary, the system improved reporting efficiency, information retrieval, and record accuracy while reducing the administrative effort associated with paper-based documentation. These findings illustrate the potential benefits of web-based information systems for organizations seeking to improve their record-management procedures.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both systems aim to improve information management by moving away from manual record-keeping practices. In the context of the present study, manually maintained subdivision layout maps can make it more difficult to maintain and retrieve current lot availability information. A centralized web-based platform can provide a more organized means of maintaining lot records and updating their availability statuses.

Although Tanaman et al. focused on inventory and asset management rather than residential lot monitoring, the study offers a useful reference for applying centralized database management to operational processes. Both systems seek to improve information accuracy, simplify record retrieval, and reduce the limitations associated with manual documentation. The inventory system's approach to electronic record management can therefore inform the organization and maintenance of lot information in the present project.

Overall, this local study provides a relevant reference for the development of a web-based solution intended to improve record management and operational efficiency. Its application to the present study lies in the shared use of digital information management principles, adapted to the specific requirements of monitoring residential lot availability at Golden Dragon Estate Corporation.

Development of GSO Item Borrowing and Inventory Management System

Estrella et al. [10] (2026) developed a GSO Item Borrowing and Inventory Management System for the General Services Office (GSO) of Tagoloan, Misamis Oriental. The study addressed inefficiencies associated with paper-based procedures for tracking borrowed items, maintaining inventory records, and processing approval requests. According to the researchers, manual documentation contributed to delayed transactions, inaccurate records, and difficulties monitoring organizational resources. To address these challenges, they developed a web-based system intended to digitize inventory operations and improve resource management.

The researchers implemented the system using a relational MySQL database to organize and maintain inventory information. Its functions included item borrowing requests, inventory tracking, approval workflows, and record maintenance. By replacing paper-based forms with digital transactions, the system provided a centralized platform for processing requests and managing records. This approach aimed to improve information consistency, reduce administrative effort, and make inventory information more accessible to authorized users.

An important aspect of the study was its consideration of both technical performance and usability. The system reportedly improved processing efficiency, data accuracy, and inventory management procedures. However, the researchers also identified challenges in user acceptance. Through the System Usability Scale (SUS), the study examined users' experiences with the system and found that adapting to the new interface presented difficulties because it differed from their familiar paper-based workflows. This finding highlighted that technical functionality alone does not guarantee successful adoption. An effective information system must also provide an intuitive interface that supports users' needs and established work practices.

The study demonstrated the importance of balancing technical implementation with user-centered design when developing organizational information systems. Its use of a centralized MySQL database provides a relevant example of managing records through a web-based platform, while its usability findings emphasize the need to consider users' familiarity with existing procedures during system development.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both projects aim to improve manual tracking and record management through web-based information systems. While Estrella et al. focused on inventory tracking and item-borrowing transactions, the present study focuses on monitoring residential lot availability and maintaining updated lot records. Despite their different applications, both systems require organized data management, efficient information retrieval, and an interface that users can understand and operate effectively.

Furthermore, the usability findings provide guidance for designing the present system's interface. An interactive dashboard with clearly distinguishable lot-status indicators and straightforward controls can help users understand lot information and perform monitoring tasks more efficiently. The use of MySQL also provides a relevant technical reference for organizing and maintaining records, although the suitability of the overall architecture depends on the specific requirements of the present project. Overall, this local study offers useful insights into both database-supported information management and the importance of usability in encouraging user acceptance.

PhilHealth Regional Office VIII Supply and Property Inventory Management System

Funcion et al. [11] (2019) developed a web-based Supply and Property Inventory Management System for PhilHealth Regional Office VIII to improve property monitoring and inventory management. The study addressed limitations in the organization's manual inventory procedures, which relied on paper-based records and physical documentation. These practices reportedly contributed to delayed reporting, computational errors, inconsistent records, and difficulties tracking the movement of assets across departments. To address these issues, the researchers developed an automated system designed to centralize property information and support more efficient monitoring.

The system was designed to manage the recording, tracking, and monitoring of organizational supplies and properties. Through a web-based platform and centralized database, authorized personnel could access and maintain inventory information more efficiently. The system supported inventory-related activities such as property recording, status monitoring, report generation, and inventory verification. By digitizing these procedures, the application aimed to reduce the time and effort required for routine inventory management and improve access to organizational records.

Another important objective was to improve data accuracy and reduce errors associated with manual record-keeping. Centralizing property information helped organize records and make them easier to retrieve. The system also supported the monitoring of asset availability, location, and status, giving personnel a more structured way to manage organizational resources. According to the supplied study summary, its implementation improved operational efficiency, record accessibility, and inventory control while reducing repetitive documentation tasks. These outcomes illustrate how automating routine information-management activities can support accountability and administrative decision-making.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both systems aim to improve monitoring through centralized web-based information management. While Funcion et al. focused on supplies and organizational properties, the present study concentrates on residential lot availability and the maintenance of lot records. Despite their different applications, both systems address the challenges of manual documentation and the need for accessible, accurate, and updated information.

Furthermore, the study provides a useful reference for implementing status monitoring and centralized record maintenance. In the present system, authorized personnel need to view and update lot statuses, including available, pending, and sold, according to the project's defined procedures. A centralized platform can help organize these records and reduce the difficulties associated with maintaining information across physical documents and subdivision layout maps. However, the effectiveness of such a system depends on accurate data entry, appropriate access controls, and timely updates by authorized users.

Overall, this local study provides a relevant reference for applying web-based monitoring and automated record management to organizational operations. Its contribution to the present project lies in demonstrating the value of replacing manual procedures with a structured digital platform adapted to the specific requirements of residential lot availability monitoring.

Re:SCN: A Web-Based Report Management System with RFID-Based Vehicle Monitoring for Villa Belen

Canlas et al. [12] (2022) developed Re, a web-based report management system integrated with radio-frequency identification (RFID)-based vehicle monitoring for Villa Belen Subdivision in Angeles City, Pampanga. The study addressed administrative challenges associated with traditional record-keeping and monitoring procedures in residential communities. According to the researchers, the absence of a centralized digital platform contributed to communication delays, inefficient monitoring, and difficulties managing community-related records and activities. These challenges highlighted the need for a more organized and accessible approach to subdivision administration.

To address these concerns, the researchers developed a web-based platform that consolidated administrative records, reporting functions, and monitoring activities. The system integrated report management with RFID-based vehicle monitoring, allowing authorized personnel to access records and monitor relevant subdivision activities through a centralized interface. By reducing reliance on manual documentation, the system aimed to improve information accessibility and streamline administrative procedures.

The study also examined the system's contribution to communication and operational efficiency. Based on the supplied study summary, user feedback was gathered through a standardized Likert-scale assessment. Respondents reportedly expressed positive evaluations of the system's information accessibility, administrative efficiency, and transparency. These findings suggest that a centralized web-based platform can support the organization of records and help administrative personnel manage routine tasks more effectively.

However, the strength of these conclusions depends on the evaluation criteria, respondent group, and results reported in the original publication.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both projects address information-management needs within residential developments. While Re focuses on report management and RFID-based vehicle monitoring, the present study focuses on maintaining residential lot records and monitoring lot availability. Despite their differences in functionality, both systems use web-based information management to make relevant records more accessible and support subdivision operations.

Furthermore, Re provides a local example of applying digital technologies to improve administrative processes within a residential community. Its centralized approach offers a useful reference for organizing lot information and making updated records accessible to authorized personnel. For the present study, a digital platform with clearly displayed lot statuses can help address the limitations of manually maintained subdivision layout maps and make lot availability information easier to retrieve and update.

Overall, this study provides a relevant local reference for the use of web-based systems in residential subdivision administration. Although its specific monitoring functions differ from those of the present project, it supports the broader principle of using centralized digital platforms to improve information accessibility and administrative efficiency. The present study applies this principle to the more specific task of monitoring residential lot availability at Golden Dragon Estate Corporation.

Vehicle Tracking and Monitoring: A Security System for Exclusive Subdivision

Lino and Festijo [13] (2019) developed a Vehicle Tracking and Monitoring System to improve security and monitoring operations within an exclusive subdivision in the Philippines. The study addressed limitations in traditional monitoring practices, which relied on manual procedures and fragmented records. These practices made it difficult for subdivision administrators to obtain timely and accurate information about vehicle movements and security-related activities. To address these challenges, the researchers developed a web-based monitoring platform intended to improve information accessibility and administrative oversight.

The system provided administrators with a centralized dashboard for viewing vehicle-monitoring information. It also stored monitoring data in a database, allowing users to access historical records and review previous activities when needed. By organizing monitoring information within a web-accessible platform, the system aimed to reduce delays associated with manual reporting and provide a more structured approach to subdivision security management.

An important feature of the study was its emphasis on presenting updated information through an interactive dashboard. This approach enabled administrators to view monitoring data in an organized format and supported timely responses to security-related concerns. The researchers also evaluated the system using software quality

criteria, including functionality, usability, and operational efficiency. According to the supplied study summary, the evaluation indicated that the system provided useful monitoring information through an accessible interface. These findings highlight the potential value of combining centralized data storage with effective information visualization in residential community management.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both systems use web-based monitoring to provide users with accessible and updated information within a residential property environment. However, their purposes differ. Lino and Festijo focused on vehicle movements and subdivision security, whereas the present study focuses on monitoring residential lot availability and maintaining accurate lot records.

The study's use of a centralized dashboard offers a useful reference for presenting information clearly and supporting administrative decision-making. In the present system, visual indicators for lot statuses, such as available, pending, and sold, can help authorized personnel identify the status of individual lots more easily. Similarly, maintaining lot information in a centralized database can support consistent record management and allow users to retrieve relevant information when needed.

Although the two systems serve different operational needs, both demonstrate the relevance of web-based platforms for monitoring information and supporting administrative activities. The study therefore provides a useful local reference for the dashboard and monitoring aspects of the present project. Its principles can be applied to lot availability management while accounting for the specific requirements of Golden Dragon Estate Corporation.

eReserve: An Online Facility Reservation System of One Philippine Higher Education Institution

Vega and Generoso [14] (2025) developed eReserve, an online facility reservation system implemented at Lyceum of the Philippines University–Batangas. The study addressed operational challenges associated with manual facility reservation procedures, including scheduling conflicts, overlapping bookings, and delays in updating facility availability. These issues created uncertainty among users regarding the availability of institutional facilities and increased the administrative effort required to manage reservations.

To address these challenges, the researchers developed a web-based reservation platform supported by a MySQL database. The system provided functionality for checking facility availability, recording reservations, and updating reservation statuses. Through a centralized database, reservation information could be stored and managed in an organized manner, helping maintain consistency between booking records and facility availability.

A notable feature of eReserve was its availability validation and reservation conflict detection. According to the supplied study summary, the system checked whether a facility was available when a reservation was requested and prevented double bookings through its reservation-confirmation mechanism. The system also maintained utilization records that helped administrators monitor facility usage and support resource allocation decisions.

These functions demonstrate how automated validation and centralized data management can reduce inconsistencies associated with manual reservation processes.

The study reportedly found improvements in data accuracy, reservation processing efficiency, and user satisfaction following the implementation of the system. Providing users with access to updated facility availability also helped improve transparency and reduce uncertainty during the reservation process. Overall, the study highlights the potential of web-based systems to improve operational efficiency by automating availability checks and maintaining consistent records.

The study is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both systems focus on maintaining accurate availability information through a centralized web-based platform. While eReserve manages institutional facility bookings, the present study focuses on monitoring residential lots and maintaining their statuses, such as available, pending, and sold. Despite the difference in application, both systems require reliable records and timely status updates to help users access current information.

Furthermore, eReserve provides a useful reference for implementing database-supported availability checks and status management. Its approach to validating facility reservations illustrates the importance of checking existing records before confirming a change in availability. A similar principle can inform lot-status management by helping authorized personnel verify the current status of a lot before updating its record. However, any prevention of conflicting or simultaneous updates in the present system would depend on the validation and database controls actually implemented.

Although the systems serve different purposes, both emphasize the value of centralized information, automated validation, and accessible status records. Therefore, the eReserve study provides a relevant local reference for the database and availability-management aspects of the present project, particularly in reducing inconsistencies and improving the reliability of information used in administrative decisions.

Summary of Local Studies

The reviewed local studies consistently demonstrate the positive impact of web-based information systems on organizational operations. Researchers have found that centralized databases, automated information processing, and digital monitoring tools improve efficiency, reduce human errors, and enhance data accessibility.

Although the reviewed studies focus on different application domains, they share common objectives related to information management, record monitoring, and operational improvement. These findings provide strong support for the development of the proposed Web-Based Lot Availability Monitoring System.

2.3 Related Literature

Real-Time Inventory Automation vs. Manual Spreadsheet Tracking

Immadisetty [15] (2025) examined the operational risks and inefficiencies associated with traditional inventory management practices, particularly those that rely on manual files and spreadsheet-based tracking. The literature emphasized the importance of accurate and timely inventory information in organizations that manage numerous assets and resources. Inaccurate or outdated records can contribute to stock shortages, over-allocation, and inefficient resource management, potentially affecting operational performance and decision-making.

One of the major limitations discussed in the literature is the reliance of spreadsheet-based workflows on manual data entry and updates. Because information must be entered and maintained by users, these processes are susceptible to human error, delayed updates, and inconsistencies between records. Consequently, decision-makers may rely on outdated information when assessing asset availability. This delay between an actual change and its reflection in the records can be described as data latency, which reduces the reliability of information used for operational decisions.

To address these limitations, Immadisetty [15] highlighted the value of real-time inventory automation supported by centralized databases and synchronized data updates. These approaches allow information to be updated when transactions occur and make current records more accessible to authorized users. Automated data management can also reduce repetitive manual work, improve record consistency, and support more informed decisions by providing a clearer view of asset availability and distribution. However, the effectiveness of these improvements depends on proper system implementation, data validation, and timely input from users.

The literature is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because both contexts involve maintaining accurate records of available assets. In the present study, manually maintained subdivision maps and related records can make it difficult to identify the current status of residential lots when updates are delayed or recorded inconsistently. A centralized web-based system can address these limitations by providing an organized repository for lot information and enabling authorized personnel to maintain and retrieve updated records.

Although inventory automation generally focuses on tracking supplies or organizational assets, the same principles can be applied to residential lot availability monitoring. The proposed system focuses on recording and displaying lot statuses, including available, pending, and sold, through a centralized digital platform. When a lot record is updated, the system can reflect that change for users accessing the latest data, subject to the system's implemented validation and synchronization mechanisms.

Overall, this literature provides a conceptual foundation for replacing fragmented manual tracking practices with centralized digital record management. It supports the importance of data accuracy, timely updates, and reduced dependence on manual documentation. Applying these principles to Golden Dragon Estate Corporation can help organize lot information, improve access to availability records, and support more efficient monitoring operations.

Single Page Application (SPA) Architecture via React.js

Kothapalli [16] (2022) discussed the role of Single Page Application (SPA) architecture in modern web development, particularly its ability to provide responsive and interactive user experiences. Unlike traditional multi-page applications that commonly reload pages when users navigate or perform actions, SPAs load a main page and dynamically update its content as users interact with the application. This approach can improve navigation and make applications more responsive, particularly when users frequently access changing information.

The literature explains that SPA architecture is suitable for applications with multiple interactive components and data-driven interfaces. By managing many interface updates on the client side, an SPA can reduce unnecessary full-page reloads and provide smoother transitions between application sections. However, overall performance also depends on the efficiency of backend requests, database operations, network conditions, and frontend implementation.

Kothapalli [16] also discussed optimization techniques that can improve SPA performance, including code splitting, lazy loading, client-side caching, and centralized state management. Code splitting divides application code into smaller bundles that can be loaded when needed, while lazy loading delays the loading of certain resources until they are required. Client-side caching can reduce repeated requests for information that does not need to be retrieved again immediately, and state management helps maintain consistent information across interface components. These techniques can contribute to a more efficient application when implemented according to its requirements.

The literature also highlighted the importance of balancing responsiveness with functional reliability. Although SPA architecture can improve the user experience, developers must ensure that data remains accurate and that interface updates correspond to the latest information available from the backend. Applications that display frequently changing records therefore require appropriate data-fetching, validation, and synchronization mechanisms in addition to an interactive frontend.

This literature is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because the system requires an interface through which authorized users can view lot information, check availability statuses, and perform permitted updates. React.js, which supports component-based interface development, can be used to create reusable elements for lot-status displays, dashboards, and other system functions. SPA architecture can also allow selected interface components to update without reloading the entire page.

For example, when an authorized user changes a lot’s status, the interface can display the updated information after the change has been successfully processed by the backend. The accuracy and timeliness of that update depend on how the application communicates with the server and retrieves the latest database records. Thus, the SPA approach supports interface responsiveness, while the backend and database remain responsible for processing and maintaining the underlying lot information.

Overall, the literature provides a technical basis for considering SPA architecture and React.js in developing an interactive lot monitoring interface. Its relevance lies in supporting efficient navigation, reusable interface components, and responsive presentation of lot information. These capabilities align with the goal of making lot availability records easier for authorized personnel to view and manage.

Server-Side Engineering with Node.js and Express.js

Shkodra et al. [17] (2021) examined the performance and efficiency of backend development technologies, focusing on Node.js and its suitability for building scalable web applications. The literature discussed the increasing need for backend systems that can process multiple client requests efficiently, particularly in applications that frequently retrieve, modify, and exchange data. In web applications with several concurrent users, inefficient request handling can lead to delays and reduced responsiveness.

Node.js uses an asynchronous, event-driven architecture and a non-blocking input/output (I/O) model to manage operations such as network communication and database requests. This approach allows the server to continue processing other tasks while waiting for certain I/O operations to finish. Consequently, Node.js can be suitable for applications that handle frequent client-server communication, including dashboards, web APIs, and information monitoring systems. However, actual performance depends on factors such as application design, database efficiency, server resources, and the nature of the workload.

The literature also discussed Express.js, a lightweight web application framework commonly used with Node.js. Express.js simplifies backend development by providing mechanisms for routing requests, handling HTTP methods, and organizing API endpoints. These capabilities support the development of RESTful APIs and the implementation of Create, Read, Update, and Delete (CRUD) operations. When combined, Node.js and Express.js provide a practical foundation for building backend services that receive requests from a frontend application, process business rules, and interact with a database.

The literature indicated that Node.js and Express.js can be effective for conventional web applications that prioritize responsiveness, maintainability, and efficient handling of I/O operations. Nevertheless, selecting an appropriate backend architecture requires consideration of the application's expected workload and performance requirements. Applications that involve computationally intensive operations may require additional optimization or a different processing approach.

This literature is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because the system requires a backend to process user requests, retrieve lot records, update availability statuses, and manage authentication-related operations. Node.js and Express.js can provide API endpoints through which the React.js frontend communicates with the server. The backend can validate incoming requests, enforce application rules, and interact with the MySQL database to maintain lot information.

For example, when an authorized user requests the status of a particular lot, the backend can retrieve the corresponding record and return the information to the frontend. When a lot status is changed, the backend can validate the request and submit the update to the database. Appropriate authorization, input validation, and database transaction controls are also necessary to protect records and reduce the risk of inconsistent updates.

Overall, this literature provides a technical basis for using Node.js and Express.js in the development of the present system. Their support for API development and asynchronous request handling aligns with the system's need to manage lot information through a web-based interface. Combined with React.js and MySQL, these technologies can form a structured architecture for accessing, processing, and maintaining lot availability records.

Relational Database Management Systems (RDBMS) via MySQL

Chittayasothorn [18] (2022) discussed the fundamental principles of Relational Database Management Systems (RDBMS), emphasizing the importance of organizing information through structured relational models. In a relational database, data is stored in tables consisting of rows and columns, with relationships established between tables through defined keys and constraints. This structure allows applications to store, retrieve, update, and manage related information in an organized manner. Relational database systems are particularly useful for applications that require consistent records and clearly defined relationships among data entities.

The literature highlighted the role of Structured Query Language (SQL) in managing relational databases. SQL supports operations such as inserting, retrieving, updating, and deleting records, as well as defining database structures and relationships. Through appropriate schema design, developers can organize information into related tables and reduce unnecessary duplication. Database normalization can further improve data organization by reducing redundant information and helping maintain consistency when records are modified.

Another important consideration is transactional integrity. Chittayasothorn [18] discussed the significance of the Atomicity, Consistency, Isolation, and Durability (ACID) properties in reliable database transaction processing. Atomicity ensures that a transaction is completed as a unit or rolled back; consistency ensures that transactions preserve defined database rules; isolation governs the interaction of concurrent transactions; and durability ensures that committed changes persist. These properties support reliable data processing, but their effectiveness depends on the database configuration, storage engine, schema constraints, and correct use of transactions.

The literature also emphasizes that selecting a relational database alone is insufficient to guarantee data accuracy. Developers must establish appropriate relationships, validation rules, constraints, and transaction procedures to protect the integrity of stored information. A well-designed database can therefore provide a dependable foundation for applications that manage important operational records.

This literature is relevant to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation because the system requires an organized database for maintaining lot information, including lot identifiers, locations within the subdivision, and availability statuses. MySQL provides

a relational database environment in which these records can be stored and linked according to the system's data requirements. A properly designed schema can help maintain consistency and support efficient retrieval and updating of lot information.

Furthermore, relational database principles are relevant when handling lot-status changes. For example, validation rules and appropriate transaction controls can help prevent invalid status values and reduce the risk of conflicting updates. If the system supports related records, such as status histories or user activity logs, relational tables and defined relationships can also help maintain an organized record of these activities. These features should be included only where they form part of the system's actual design and implementation.

Overall, this literature provides a theoretical basis for using MySQL and relational database design in the present project. Its emphasis on structured data organization, normalization, constraints, and transaction integrity supports the development of a reliable database for residential lot availability monitoring. Applying these principles can help improve the consistency, accessibility, and maintainability of the information managed by Golden Dragon Estate Corporation.

Software Usability and Evaluation Quality Metrics

Klug [19] presented a methodological overview of the System Usability Scale (SUS), a standardized instrument originally developed by John Brooke in 1986 to evaluate users' perceived usability of software systems. SUS consists of ten questionnaire items rated using a Likert scale and measures users' overall perceptions of a system's usability, including its ease of use, complexity, consistency, and learnability. Due to its simplicity and applicability across different technologies, SUS has become a widely used instrument for evaluating the usability of websites, software applications, and interactive systems.

The use of SUS is particularly relevant to systems intended for users with varying levels of technical expertise. By converting users' subjective experiences into a numerical score, the instrument provides a standardized basis for assessing perceived usability and comparing results across different systems or development iterations. The resulting feedback can help developers identify potential usability problems and determine areas that require interface improvements. In the context of the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*, SUS can provide a structured means of evaluating whether real estate personnel can navigate the system, interpret lot information, and perform lot-status management tasks efficiently.

Similarly, Awangditama et al. [20] discussed the importance of applying standardized software quality models, particularly ISO/IEC 25010, to assess the quality of software systems. This framework organizes software quality into defined characteristics that can guide the evaluation of a system's technical capabilities and suitability for its intended use. Depending on the applicable version of the model, these characteristics may include functional suitability, performance efficiency, compatibility, interaction capability or usability, reliability, security,

maintainability, and flexibility. Applying a standardized quality model helps evaluators establish consistent criteria for determining whether a software system satisfies its intended requirements.

SUS and ISO/IEC 25010 serve complementary purposes in software evaluation. SUS primarily measures users' perceived usability through questionnaire responses, whereas ISO/IEC 25010 provides a broader framework for examining software quality characteristics. Using both approaches can support a more comprehensive assessment by considering the user experience alongside technical and operational requirements. However, the specific quality characteristics selected for evaluation should correspond to the system's actual functions, requirements, and testing procedures.

These evaluation approaches are relevant to the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation* because the system must be both understandable to its intended users and capable of managing lot information accurately and securely. SUS can be used to assess the perceived usability of the interface, including the ease with which authorized personnel can view lot availability, identify lot statuses, and update records. This is important because the system's usefulness depends partly on whether personnel can perform these activities without unnecessary difficulty.

Meanwhile, ISO/IEC 25010 can guide the evaluation of the system's technical quality. Relevant characteristics may include functional suitability, performance efficiency, reliability, security, and usability or interaction capability, as applicable to the selected model version. These characteristics are particularly important because the system manages information about lot availability and status changes, which must be presented consistently and protected against unauthorized access. The evaluation results can help determine whether the implemented system meets its functional requirements and identify areas for improvement.

Overall, the use of SUS alongside an appropriate ISO/IEC 25010 quality model provides a structured approach to evaluating both perceived usability and broader software quality. These frameworks support the assessment of whether the developed system is user-friendly, functionally suitable, reliable, and secure for its intended operational environment.

Role-Based Access Control (RBAC)

Owoade et al. [21] discussed Role-Based Access Control (RBAC) as a security model for managing user permissions in information systems. Under this model, permissions are assigned to predefined roles, and users receive access rights according to their assigned roles and responsibilities. Rather than configuring permissions individually for every user, administrators can manage access through a structured set of roles. This approach simplifies permission management and supports the consistent enforcement of organizational security policies.

RBAC is particularly useful in multi-user environments where individuals require different levels of access to system functions and information. For example, roles such as administrator, manager, and standard user may be assigned distinct permissions based on their responsibilities. By limiting access to the functions necessary for each

role, RBAC helps reduce opportunities for unauthorized data access and modification. Properly implemented role-based permissions can also reduce accidental changes and support the integrity and confidentiality of organizational information.

Another advantage of RBAC is its manageability and scalability. As an organization adds users or modifies responsibilities, administrators can assign existing roles or update role permissions rather than manually configuring every user account. This structure can reduce administrative effort and make access policies easier to maintain. Furthermore, clearly defined roles and permissions help organizations document and review who is authorized to perform particular operations. However, RBAC must be supported by appropriate authentication, server-side authorization checks, and regular permission reviews to provide effective protection.

The principles of RBAC are relevant to the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*. Since the system manages lot information and availability statuses, access to its functions must be restricted according to users' responsibilities. For instance, real estate personnel may be permitted to view lot availability and search for lot information, while authorized administrators may be allowed to create, update, or delete records and manage user accounts, depending on the system's approved requirements.

Separating these permissions helps prevent unauthorized users from making changes to important lot records and supports the consistency of the information presented by the system. The role-based structure can also simplify account administration when additional personnel or responsibilities are introduced. Nevertheless, the actual permissions assigned to each role must reflect the organization's operational policies and the functions implemented in the system.

Overall, RBAC provides a structured approach to managing access permissions in multi-user web applications. Its application to the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation* supports the implementation of controlled access to lot information and administrative functions. When combined with secure authentication and server-side permission validation, RBAC can help strengthen system security and protect the integrity of lot availability records.

Agile Software Development Framework

Irfansyah et al. [22] discussed the importance of Agile software development methodologies, particularly the Scrum framework, in managing software projects with changing requirements. Agile emphasizes iterative development, collaboration, continuous feedback, and the incremental delivery of functional software. These principles are useful in web-based application development, where user needs and operational requirements may change throughout the development process. Compared with a strictly linear approach, iterative development provides opportunities to identify issues earlier and adjust system features as new requirements emerge.

Scrum organizes development work into time-limited cycles known as sprints. During each sprint, the development team works toward a defined objective and produces an increment of the system that can be reviewed and tested. Depending on the project's scope, these increments may include user authentication, data management functions, interface components, or dashboard features. Sprint reviews and stakeholder feedback allow the team to evaluate completed work, identify necessary improvements, and determine priorities for subsequent development activities.

Another important principle of Agile development is collaboration among developers, stakeholders, and intended users. Regular communication helps the team clarify requirements, evaluate proposed features, and address usability or functional issues before they become more difficult to resolve. This feedback-driven approach is particularly valuable for applications designed to support organizational workflows because the system must reflect actual operational needs rather than relying solely on initial assumptions.

Agile development also supports adaptability when requirements evolve during implementation. By reviewing and refining system components incrementally, developers can make adjustments to functionality and interface design throughout the development lifecycle. However, the effectiveness of this approach depends on clear development objectives, active stakeholder participation, appropriate testing, and consistent documentation. Agile practices do not automatically guarantee software quality; each increment must still be evaluated against the project's requirements.

The literature on Agile software development is relevant to the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation* because the application must address the information-management needs of real estate personnel. Features such as lot-status management, user authentication, lot information displays, and dashboard visualization may require refinement as users review the system and provide feedback. An iterative development approach can help ensure that these functions are tested and improved in accordance with the organization's actual workflows.

Furthermore, stakeholder involvement can help determine whether the digital presentation of lot availability is understandable and whether the available system functions support personnel in managing lot records. Feedback collected during development and testing can guide adjustments to the interface, validation procedures, and system functionality. This is especially relevant when replacing or supplementing manual, color-coded lot maps with a digital monitoring platform.

Overall, Agile software development provides a flexible approach to organizing software development, testing, and refinement. Its emphasis on incremental progress and stakeholder feedback offers a useful methodological foundation for developing a system that responds to operational requirements. If Scrum was the framework actually followed by the project team, its sprint-based practices can also explain how the system's modules were developed, reviewed, and improved before deployment.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign studies, local studies, and related literature provide a foundation for the design and development of web-based information systems intended to improve information management, operational efficiency, and data accessibility. Collectively, the reviewed works demonstrate the value of replacing or improving manual record-keeping processes through computerized systems that support centralized data storage, organized information retrieval, and more consistent record management. They also identify common challenges associated with manual processes, including delayed updates, data inconsistencies, human error, and difficulties in retrieving and maintaining records.

The foreign studies primarily examine web-based real estate, property management, and inventory systems that demonstrate the application of modern software technologies and database-driven architectures. These studies emphasize the importance of organized data management, accessible interfaces, appropriate security controls, and efficient processing of information. The related literature on React.js, Node.js, Express.js, and MySQL further explains how frontend technologies, backend services, and relational databases can support the development of interactive and maintainable web applications. However, the use of these technologies alone does not guarantee real-time updates, system security, or data integrity; these capabilities also depend on the application's implementation, database design, and access-control mechanisms.

The local studies provide additional context by examining information systems implemented in Philippine organizations and operational environments, including property management, inventory management, vehicle monitoring, and facility reservation. These studies demonstrate how computerized systems can support more organized record management, information retrieval, monitoring, and administrative processes. They also provide lessons regarding usability, workflow alignment, and the importance of evaluating whether a system addresses the actual needs of its intended users. Although these studies concern different operational domains, their findings and approaches offer useful guidance for developing an information system suited to the requirements of Golden Dragon Estate Corporation.

The related literature strengthens the conceptual and technical foundation of the present study by discussing database-driven automation, single-page application architecture, backend development, relational database management, Role-Based Access Control (RBAC), Agile software development, and software quality evaluation. These concepts address complementary aspects of system development. Centralized database management supports organized and consistent records; appropriate frontend and backend implementation supports user interaction and information processing; RBAC provides a structured approach to controlling access permissions; and iterative development can help refine system functions based on feedback. Meanwhile, the System Usability Scale (SUS) and the applicable ISO/IEC 25010 software quality model can guide the evaluation of perceived usability and relevant software quality characteristics.

Taken together, the reviewed works suggest that effective web-based information systems require more than the digitization of existing records. They must also provide appropriate data management, usable interfaces, controlled access to sensitive functions, and evaluation procedures that determine whether the system meets its

intended requirements. For lot availability monitoring, these considerations are particularly important because users need to identify individual lots, understand their current availability statuses, and access reliable information when performing real estate operations.

The present study applies these insights to the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. In contrast to systems focused primarily on general property listings, inventory records, reservations, or asset tracking, the present system centers on monitoring residential lot availability through a visual representation of the subdivision layout and associated lot-status information. Its purpose is to organize lot records in a centralized web-based platform and help authorized personnel view and manage availability information more efficiently. The reviewed literature therefore provides guidance for the system's architecture, data management, access control, interface design, development approach, and evaluation.

Research Gap

Despite the availability of foreign and local studies on web-based real estate management, inventory management, reservation, and monitoring systems, a specific application need remains relevant to the present study: the integration of centralized lot information management with an intuitive, map-based visualization of residential lot availability for the operational requirements of a Philippine real estate company.

Many of the reviewed systems address broader information-management needs, such as maintaining property listings, managing inventory records, processing reservations, monitoring vehicles, or administering organizational assets. Although these systems provide useful approaches to database management, access control, information retrieval, and system evaluation, their primary objectives differ from the specific task of visually monitoring the availability status of individual residential lots within a subdivision layout. Consequently, their approaches may not directly address the workflow of personnel who need to locate a particular lot on a visual map and determine whether it is available, pending, or sold.

Another consideration is the coordination of lot-status information with its visual representation. When lot information is maintained through manual records or color-coded subdivision maps, personnel may need to consult multiple sources or update information separately. Such processes can create opportunities for inconsistent records, delayed status changes, and errors in communicating availability. A web-based platform that links lot records to a visual subdivision layout can help address these challenges by providing a more organized way to view and manage lot information. The effectiveness of this approach, however, depends on appropriate database design, validation, access permissions, and procedures for maintaining accurate records.

The reviewed literature also discusses several important development and evaluation approaches, including modern web technologies, RBAC, Agile development, SUS, and ISO/IEC 25010. These approaches contribute to

different aspects of software development and quality assurance, but their individual discussion does not by itself establish that they have been integrated into a system specifically tailored to the lot-monitoring workflow of Golden Dragon Estate Corporation. The relevant gap is therefore not the absence of web-based information systems in general, but the need to address the organization's specific lot-availability monitoring requirements through a unified application.

To address this need, the Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation focuses on providing a centralized platform for viewing and managing residential lot availability through a visual subdivision layout and associated status records. The system is intended to support more organized information retrieval, consistent lot-status management, and controlled access to relevant functions. Its usability and software quality can also be evaluated using appropriate assessment instruments and criteria. Through this focused application, the study seeks to provide a practical alternative to fragmented or manual lot-monitoring procedures and support more efficient real estate operations.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study employed a developmental research design to design, develop, and evaluate the *Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*. Developmental research focuses on the systematic development and evaluation of a product, system, or tool intended to address a specific practical problem. This design was appropriate for the study because its primary objective was to develop a functional web-based application that would improve the company's existing process of monitoring residential lot availability.

The study addressed the limitations associated with manual lot monitoring by developing a centralized platform for managing and viewing lot availability information. The system was designed to support more organized record management, accessible lot information, and consistent monitoring of lot statuses. Its development considered the operational requirements of Golden Dragon Estate Corporation, particularly the need to identify individual lots and manage their availability information through a digital platform.

The developmental process involved identifying the existing operational concerns, gathering system requirements, designing the system architecture and database, developing the application, and evaluating the resulting system against its intended requirements. These activities guided the development of the system's interface, lot information management functions, and access controls. The evaluation process was intended to determine whether the implemented system met its functional requirements and provided an appropriate level of usability and software quality for its intended users.

Furthermore, the study incorporated an iterative approach to system improvement by allowing feedback from users and stakeholders to inform revisions to the system's functions and interface. This approach supported the refinement of the application in response to identified issues and operational needs. Where applicable, these iterative practices were aligned with Agile development principles, which emphasize incremental development, collaboration, and continuous improvement.

Overall, the developmental research design provided a structured basis for producing and evaluating a web-based solution to the company's lot availability monitoring needs. It connected the identification of the existing problem with the development of a practical system and its subsequent evaluation, ensuring that the project remained focused on the operational requirements of Golden Dragon Estate Corporation.

3.2 System Development Methodology

The study employed the Agile Software Development Methodology as a general approach to developing the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*. Agile is an iterative and incremental approach to software development that emphasizes adaptability, collaboration, feedback, and continuous improvement. It allows system requirements and features to be refined throughout development based on identified issues and user needs.

The Agile approach was appropriate for the study because the system needed to address the operational requirements of Golden Dragon Estate Corporation while allowing improvements to its functions and user interface during development. Through an iterative process, the researchers developed the system components, tested their functionality, and incorporated feedback to improve the application before deployment.

The system development process was organized into the following phases: planning, analysis, design, and development and implementation. These phases provided a structured process for identifying the company's needs, defining system requirements, preparing the system design, and developing the web-based application.

3.2.1 Planning Phase

The Planning Phase involved identifying the objectives, scope, and initial requirements of the system. During this phase, the researchers conducted initial discussions and observations at Golden Dragon Estate Corporation to understand the company's existing process for monitoring residential lot availability.

The researchers identified that the company used printed subdivision maps with manually applied color codes to indicate lot availability. Green represented available lots, yellow represented pending lots, and red represented sold lots. Although this method provided a visual means of identifying lot statuses, it relied on manual updates and could lead to inconsistencies, delays, and difficulties in maintaining accurate information.

Based on the initial observations, the researchers established the main objectives of the system as follows:

- Automating the recording and updating of lot availability statuses.
- Providing an interactive visual interface for monitoring individual lots.
- Centralizing lot information in a database.
- Restricting access to system functions through user authentication and appropriate permissions.
- Improving the accessibility and organization of lot availability information.
- Supporting relevant reporting and email notification functions, where required by the system's operational scope.

The researchers also selected the technologies used in developing the application:

- **React** for frontend development.
- **Node.js and Express.js** for backend development and request processing.
- **MySQL** for database management.
- **Tailwind CSS** for user interface styling and responsive layouts.
- **Nodemailer with SMTP** for sending email notifications.

These technologies were selected to support the development of a web-based application with an interactive interface, centralized data storage, and organized communication between the frontend, backend, and database.

3.2.2 Analysis Phase

The Analysis Phase focused on examining the existing lot-monitoring process and identifying the functional and non-functional requirements of the system. The researchers analyzed the limitations of the manual approach and determined the system capabilities needed to support the company's lot availability monitoring activities.

The analysis identified the following limitations of the existing process:

- Delays in updating lot availability information.
- The possibility of human error when recording or changing lot statuses.
- Difficulty in quickly identifying the availability of individual lots.
- Dependence on physical subdivision maps and manually maintained records.
- Risks of damage, loss, or inconsistent updates to physical records.
- Limited centralized access to updated lot information.

Based on these findings, the researchers established the functional and non-functional requirements of the system.

Functional requirements

The functional requirements specified the operations that the system was designed to perform:

- **User authentication:** Allow authorized users to log in and log out of the system.
- **Lot status management:** Allow authorized personnel to record and update lot statuses.

- **Interactive lot viewing:** Display the subdivision layout and use color-coded indicators to identify lot availability.
- **User management:** Support the management of user accounts and permissions by authorized administrators.
- **Data storage and retrieval:** Store and retrieve lot and user information from the centralized database.
- **Email notifications:** Send relevant email notifications through the configured SMTP service.
- **Reporting:** Generate or display reports related to lot information and availability, according to the system's implemented functions.

Non-functional requirements

The non-functional requirements describe the expected quality and operational characteristics of the system

- **Security:** Protect user accounts and restrict access to authorized functions.
- **Performance efficiency:** Process requests and display information within an acceptable response time.
- **Usability:** Provide a clear and understandable interface for the intended users.
- **Reliability:** Maintain consistent system behavior during normal operation.
- **Data integrity and accuracy:** Support accurate storage, retrieval, and updating of lot information.
- **Availability:** Make the application accessible when the system environment and hosting infrastructure are operational.

These requirements served as the basis for the system design, implementation, and subsequent evaluation

3.2.3 Design Phase

The Design Phase involved preparing the system's technical structure, database organization, information flows, and user interface before and during implementation. The researchers developed the system design artifacts to represent how the application would process information, store records, and support interactions between users and the system.

The design activities included preparing the following:

- System architecture design.
- Database schema.
- Entity-Relationship Diagram (ERD).
- Data Flow Diagrams (DFDs).
- Use Case Diagram.

- User interface mockups.

The system was designed as a **web-based client-server application** consisting of three main components.

- **Frontend:** React handled the user interface and interactions, while Tailwind CSS supported the styling and layout of the application.
- **Backend:** Node.js and Express.js handled server-side processing, API requests, and communication between the frontend and database.
- **Database:** MySQL stored the system's structured information, including user accounts, lot records, and other data required by the implemented functions.

The user interface was designed to make lot availability information easier to understand. A central feature was the interactive subdivision layout, which used color-coded indicators to represent lot statuses:

- **Green — Available**
- **Yellow — Pending**
- **Red — Sold**

The visual indicators allowed users to identify lot statuses more quickly than by relying exclusively on text-based records. The design also supported access to relevant lot information, subject to the functions and permissions provided by the application.

In addition to the interface, the database design organized the system's records to support information retrieval and updates. The ERD represented the relationships among the database entities, while the DFDs illustrated the movement of information through the system. The Use Case Diagram described the interactions between system actors and the functions available to them. Together, these design artifacts provided a reference for implementation and helped align the application with the identified requirements.

3.2.4 Development and Implementation Phase

The Development and Implementation Phase involved translating the approved system design into a functioning web-based application. Following the iterative approach adopted in the study, the researchers developed the system components, tested their functions, and made revisions when issues or improvement opportunities were identified.

The system was organized into the following functional modules:

- **Authentication Module:** Managed user login and logout and supported access to the application by authorized users.

- **Lot Management Module:** Supported the recording, viewing, and updating of lot information and availability statuses.
- **User Management Module:** Supported administrative management of user accounts and their assigned access permissions.
- **Dashboard Module:** Presented relevant lot availability information through the system interface.
- **Email Notification Module:** Supported the sending of relevant notifications through Nodemailer and the configured SMTP service.
- **Reporting Module:** Supported the generation or presentation of reports available in the implemented system.

The frontend was developed using React and Tailwind CSS to provide an interactive interface and organize the presentation of lot information. The backend was developed using Node.js and Express.js to process application requests and coordinate communication between the frontend and the MySQL database. MySQL was used to store and manage the structured records required by the system.

During implementation, the researchers integrated the modules according to their functional requirements. The application was checked to determine whether its components performed their intended operations and whether information was processed and displayed appropriately. Feedback from users, stakeholders, and group members was considered when identifying areas for revision and improvement.

The iterative development process supported the refinement of system functions and interface elements as development progressed. Where changes were necessary, the researchers revised the relevant components and checked their behavior before proceeding with further integration. These activities helped align the implemented application with the operational requirements of Golden Dragon Estate Corporation.

Upon implementation, the resulting application provided a centralized web-based platform for monitoring and managing lot availability information. Its final quality and suitability were to be assessed through the testing and evaluation procedures described in the succeeding sections of this chapter.

3.2.5 Testing Phase

The Testing Phase was conducted to verify whether the developed system performed its intended functions and met the identified system requirements. Testing was carried out to identify errors, examine interactions between system components, and determine whether the application operated as expected.

The following testing activities were considered:

- **Functional Testing:** Verified whether the system's features, such as user authentication, lot-status management, lot viewing, and data retrieval, performed according to their intended functions.
- **Integration Testing:** Examined whether the frontend, backend, database, and other implemented modules communicated and operated correctly together.
- **Interface Testing:** Checked the usability and behavior of interface elements, including navigation menus, forms, buttons, dashboards, and the interactive subdivision layout.
- **Security Testing:** Examined authentication and access-control mechanisms to determine whether unauthorized users were restricted from protected functions.

Errors and defects identified during testing were corrected, and the affected functions were checked again to determine whether the corrections resolved the issues. This process supported the refinement of the application and helped ensure that its major components operated consistently.

Testing was incorporated into the iterative development process, allowing the researchers to identify and address issues as system components were developed and integrated. The results of testing also provided a basis for determining whether the system was ready for deployment and further evaluation.

3.2.6 Deployment and Maintenance Phase

T**he** Deployment and **Maintenance** Phase covered the preparation of the developed system for operational use and the activities necessary to keep it functional after deployment. The system was deployed for use by Golden Dragon Estate Corporation and made accessible to authorized users through a web browser, subject to the availability of the hosting environment and network connection.

Deployment involved configuring the application and its database in the designated server environment and checking whether the system could be accessed and its major functions operated as intended. User accounts and permissions were managed according to the access requirements established for the organization. Where applicable, users were provided with guidance on accessing the system, viewing lot availability, and performing the functions assigned to their roles.

Maintenance activities were intended to support the continued operation and improvement of the system. These activities include:

- **Bug Fixing:** Identifying and correcting errors discovered during actual use.
- **Performance Optimization:** Improving inefficient operations when necessary.
- **Security Updates:** Applying relevant updates and reviewing access controls to help protect the system.

- **Data Maintenance:** Checking the accuracy and consistency of lot records and maintaining appropriate backups where supported.
- **Feature Enhancements:** Considering user feedback and operational requirements when determining possible system improvements.

These activities help maintain the usefulness, reliability, and security of the application over time. Maintenance requirements may change as the organization's operational needs, software dependencies, and technical environment evolve.

3.3 System Architecture

The *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation* follows a client-server architecture supported by a database layer. This architecture separates the user interface from server-side processing and data storage, allowing each component to perform a defined role within the application.

The frontend provides the interface through which users access the system, while the backend processes requests and applies the application's business rules. The database stores the structured information required for lot monitoring and related system functions. Communication among these components allows authorized users to retrieve lot information and submit changes through the web interface.

3.3.1 Overview of the System

The Web-Based Lot Availability Monitoring System was developed to provide a digital platform for viewing and managing residential lot availability information. It supports the company's transition from manually maintained, color-coded subdivision maps to a centralized web-based application.

When a user performs an operation, such as viewing a lot or submitting a status change, the frontend sends the corresponding request to the backend. The backend processes the request, checks the applicable rules and permissions, and communicates with the MySQL database when data retrieval or modification is required. The server then returns an appropriate response to the frontend, which displays the resulting information to the user.

For example, when an authorized user updates a lot's status, the application processes the request and records the change in the database if the operation is valid and permitted. The interface can then display the updated status after receiving the server's response. The exact timing of this update depends on the application's implementation and communication process.

The system uses color-coded indicators to help users identify lot availability:

- **Green — Available**
- **Yellow — Pending**

- **Red — Sold**

These visual indicators provide a straightforward way to interpret the status of individual lots. The system's ability to display accurate information depends on valid records, appropriate access controls, and timely updates by authorized users.

3.3.2 Client-Server Setup

The system consists of three primary components: the client-side frontend, the server-side backend, and the database layer.

Client Side (Frontend)

The frontend was developed using React and Tailwind CSS. It provides the interface through which users interact with the system. Its responsibilities include:

- Displaying the interactive subdivision layout.
- Presenting the dashboard and relevant lot information.
- Providing login forms and other implemented user interface pages.
- Accepting user input through forms and controls.
- Displaying lot-status information and server responses.

Server Side (Backend)

The backend was developed using Node.js and Express.js. It handles requests from the frontend and processes application operations. Its responsibilities include:

- Receiving and processing API requests.
- Applying server-side business rules and validation.
- Handling authentication and authorization, according to the implemented security mechanisms.
- Communicating with the MySQL database.
- Returning responses to the frontend.
- Coordinating email notifications when the corresponding feature is enabled.

Database Layer

The system uses MySQL as its relational database management system. The database stores structured information required by the application, which may include:

- User account information.
- Lot details and availability statuses.
- Subdivision or project information, if applicable.
- Other records required by the implemented functions.

The database provides a centralized location for storing and retrieving system information. Appropriate table relationships, constraints, validation, and database operations help maintain the consistency and integrity of stored records.

3.3.3 User Interaction Flow

The following steps describe the general interaction between users, the application, and the database during lot monitoring and status updating.

1. The user opens the web application and enters their login credentials.
2. The backend processes the login request and verifies the credentials using the implemented authentication mechanism.
3. If authentication is successful, the user is granted access to the functions permitted by their account and role.
4. The frontend displays the dashboard and the available lot-monitoring interface.
5. The user selects a lot from the subdivision layout to view its information.
6. If authorized, the user submits a status change, such as changing a lot's status to available, pending, or sold.
7. The backend receives the request and validates the submitted information and the user's permission to perform the operation.
8. If the request is valid and authorized, the backend updates the relevant record in the MySQL database.
9. The server returns a response, and the frontend displays the resulting lot information or an appropriate error message.

10. If the operation triggers an implemented email notification, the application sends the corresponding notification through the configured email service.

This interaction flow illustrates how the frontend, backend, and database work together to support lot availability monitoring. It also highlights the importance of validating user requests and checking permissions before modifying important lot records.

3.4 Tools and Technologies Used

The development of the *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation* involved web development technologies and tools used to build, style, process, and store application data. The technology stack included JavaScript, React, Tailwind CSS, Node.js, Express.js, and MySQL, with Nodemailer and SMTP used for email functionality where applicable.

Each technology served a specific purpose in the development process, from creating the user interface to processing application requests and managing structured records.

3.4.1 Programming Language

JavaScript

JavaScript is a programming language widely used in web development to create interactive user interfaces and implement server-side application logic. It supports event handling, data processing, and asynchronous operations that allow web applications to communicate with backend services.

In this study, JavaScript was used in two main areas:

- **Frontend development:** JavaScript was used with React to implement user interactions and manage the presentation of system information.
- **Backend development:** JavaScript was used with Node.js and Express.js to process requests and implement server-side operations.

Using JavaScript across the frontend and backend allowed the development team to work with a common programming language across major application components. However, data consistency, security, and performance also depended on the system's architecture, validation procedures, and implementation.

3.4.2 Front-End Library

React

React is an open-source JavaScript library for building user interfaces through reusable components. It supports the development of interactive web applications by allowing developers to organize interface elements into manageable components and update displayed content in response to application state changes.

In the system, React was used to develop interface components such as:

- The interactive subdivision layout.
- The dashboard.
- Login and other implemented account pages.
- Forms for viewing and updating lot information.
- Navigation elements and status indicators.

React's component-based approach helped organize the interface and reuse common elements across different parts of the application. Its rendering mechanisms also supported updating interface components when the relevant application state changed, without requiring a full-page reload for every interaction.

However, React does not independently guarantee real-time synchronization with the database. The display of updated lot information depends on the application's communication with the backend and the method used to retrieve or refresh data.

3.4.3 Styling Framework

Tailwind CSS

Tailwind CSS is a utility-first CSS framework that provides utility classes for styling web interfaces. It allows developers to apply layout, spacing, typography, color, and responsive design properties directly within the application's markup or JSX.

In this study, Tailwind CSS was used to:

- Create and organize responsive layouts.
- Style buttons, forms, and navigation elements.
- Maintain consistent visual styling across interface components.
- Present lot-status indicators using the designated colors.
- Improve the readability and organization of the user interface.

The framework supported the creation of a consistent interface while reducing the need to write repetitive custom CSS. The resulting usability of the application nevertheless depended on the overall interface design and how effectively the functions were presented to users.

3.4.4 Back-End Technologies

Node.js

Node.js is a JavaScript runtime environment that allows JavaScript code to execute outside a web browser. It uses an event-driven architecture and supports non-blocking input/output operations, making it suitable for many web servers and API-based applications.

In the system, Node.js provided the runtime environment for the backend. It supported:

- Executing server-side JavaScript.
- Processing application operations.
- Supporting communication with the database through the application's database-access components.
- Handling asynchronous operations and API-related tasks.
- Supporting authentication and authorization logic implemented in the backend.

Node.js provides capabilities for handling concurrent operations, but actual performance depends on the application's implementation, database operations, server resources, and workload.

Express.js

Express.js is a web application framework for Node.js that simplifies the development of web servers and APIs. It provides routing and middleware mechanisms for processing HTTP requests and returning responses.

In this study, Express.js was used to support:

- Defining API endpoints.
- Managing routes between frontend requests and backend operations.
- Processing HTTP requests and responses.
- Organizing server-side application logic.
- Applying middleware for validation, authentication, or other implemented requirements.

Express.js helped organize the backend operations and communication between the frontend and database. Security depended on how the application implemented authentication, authorization, input validation, and other relevant controls.

3.4.5 Database Management System

MySQL

MySQL is a relational database management system that stores structured information in tables and supports data retrieval and modification through SQL. It provides mechanisms for defining relationships, constraints, and database operations according to the application's data requirements.

In the system, MySQL was used to store and manage information required for lot availability monitoring, including:

- User account information.
- Lot details and status records.
- Subdivision or project information, if included in the database design.
- Other records required by the implemented modules.

The relational structure allowed the application to organize records and retrieve relevant information through database queries. Appropriate schema design, primary and foreign keys, validation, and transaction handling helped support data integrity and consistency.

Using a centralized database also allowed the application to access a common set of lot records rather than relying solely on separately maintained physical maps. The accuracy of displayed information still depended on correct data entry, valid updates, and appropriate database and application logic.

3.4.6 Email Notification Service

Nodemailer and SMTP

Nodemailer is a Node.js module used to send email messages from applications. It can connect to an SMTP service to transmit messages through a configured email account or provider.

In the system, Nodemailer and SMTP were used to support email notifications for the functions enabled in the application. Depending on the implemented configuration, these may include:

- Account-related email messages.
- Password recovery messages.
- System alerts.
- Notifications associated with lot-status changes.

Only the email functions actually implemented and tested should be included in the final list. The successful delivery of email notifications depends on correct SMTP configuration, valid recipient addresses, provider requirements, and network availability.

Email notifications can supplement the application's interface by informing users about relevant events. They do not, by themselves, guarantee that a message is delivered immediately or read by its recipient.

3.4.7 Code Editor and Development Tools

Visual Studio Code (VS Code)

Visual Studio Code is a source code editor developed by Microsoft. It provides features for editing code, navigating project files, using extensions, and running development tools through an integrated terminal.

During development, VS Code was used to support activities such as:

- Writing and editing source code.
- Organizing and managing project files.
- Identifying syntax errors and debugging application issues.
- Running development commands through the integrated terminal.
- Using available extensions to support the development workflow.

The editor provided a working environment for managing the source code and supporting the development process.

3.4.8 Version Control System

Git and GitHub

Git is a distributed version control system that records changes to files and source code over time. GitHub is a hosting platform that supports Git repositories and collaborative software development.

Where used by the development team, Git and GitHub supported the following activities:

- Tracking changes to source code.
- Maintaining a history of development revisions.
- Coordinating code contributions among team members.
- Supporting collaboration through shared repositories.

- Recovering earlier versions of files when necessary.

Version control can help teams manage changes and review earlier versions of the project. However, repository hosting should not be treated as a substitute for a separate backup strategy or secure management of sensitive credentials and configuration files.

3.4.9 Summary of Tools and Technologies

The combination of JavaScript, React, Tailwind CSS, Node.js, Express.js, and MySQL provided the principal technologies used to develop the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Each component served a distinct role in the application: React and Tailwind CSS supported the frontend, Node.js and Express.js supported server-side processing, and MySQL provided structured data storage. Nodemailer and SMTP supported the email functions implemented in the system.

Development tools such as Visual Studio Code supported code editing and debugging, while Git and GitHub supported version tracking and collaboration where used by the team.

Together, these technologies provided the foundation for a web-based application designed to centralize lot information and present lot availability through a visual interface. The effectiveness, security, reliability, and usability of the resulting system depended not only on the selected technologies but also on the quality of the implementation, testing, database design, and operational procedures.

3.5 Data Gathering Procedures

The data-gathering procedures were conducted to obtain information about the existing lot monitoring process, operational challenges, user requirements, and organizational needs of Golden Dragon Estate Corporation. The information gathered served as the basis for identifying system requirements and determining the functions needed in the Web-Based Lot Availability Monitoring System.

The researchers used interviews, observation, document analysis, and requirements validation to understand the company's existing workflow. These methods provided information about how lot availability was monitored, how records were maintained, and what improvements were needed to support more efficient information management.

The information gathered was used during system analysis, database design, interface development, and the identification of functional and non-functional requirements. Using multiple data-gathering methods also allowed the researchers to compare information from different sources and develop a more complete understanding of the organization's operational needs.

3.5.1 Interview Method

The interview method was used to gather information from personnel involved in monitoring lot availability, maintaining subdivision records, and handling related real estate operations. The interviews aimed to understand the existing procedures, identify operational difficulties, and determine the expectations of the intended users.

The researchers prepared guide questions concerning the current lot monitoring process, the recording of lot statuses, the retrieval of lot information, and the challenges encountered when maintaining physical records. The questions also addressed the functions that users considered necessary for a web-based monitoring system.

Based on the information gathered, the existing process involved using printed subdivision layout maps and color-coded indicators to identify lot availability. Green represented available lots, yellow represented pending lots, and red represented sold lots. Although this approach was familiar to personnel, maintaining and updating the records could become difficult when managing multiple lots or subdivision projects.

The concerns identified during the interviews included the following:

- Delays in updating lot status information.
- Difficulty monitoring changes across subdivision projects.
- Risk of human error during manual updates.
- Possibility of physical records being damaged or misplaced.
- Limited access to lot information when records were unavailable.
- Lack of a centralized platform for maintaining lot monitoring information.

The information gathered also helped identify the need for a system that would allow authorized personnel to view lot information, update statuses through a digital interface, and access centralized records. These findings contributed to defining the system requirements and aligning its functions with the operational needs of Golden Dragon Estate Corporation.

3.5.2 Observation Method

The observation method was used to understand how lot monitoring activities were performed within the organization. This method allowed the researchers to examine the existing workflow and compare actual work practices with the information gathered through interviews.

The observation focused on the procedures for locating lots, updating lot statuses, maintaining subdivision layout maps, and retrieving information in response to inquiries. Particular attention was given to how employees recorded changes and ensured that the information reflected the current status of each lot.

The existing manual process involved locating individual lots on printed maps and updating their statuses through color-coded markings or annotations. Maintaining consistent records required careful handling of the maps and coordination when changes occurred. Retrieving information could also require searching through physical records.

The observation helped identify the operational concerns associated with manual monitoring, including:

- Physical wear and tear of printed maps.
- Risk of misplacing important documents.
- Difficulty maintaining consistent and updated records.
- Delays in locating specific lot information.
- Limited accessibility when physical records were unavailable.

The information obtained through observation supported the analysis of the existing workflow and helped determine how the proposed system could provide a more organized method of displaying and maintaining lot availability information. The observations were also compared with the interview findings to identify common concerns and requirements.

3.5.3 Document Analysis

Document analysis was used to examine the records and materials relevant to lot availability monitoring at Golden Dragon Estate Corporation. This method helped the researchers understand the types of information maintained by the organization and determine which data fields were necessary for the system.

Depending on the records made available for review, the documents examined included printed subdivision layout maps, lot inventory records, sales monitoring documents, reservation records, customer information forms, and internal reports related to lot status monitoring.

The analysis helped identify information that could be required by the system, such as:

- Lot number.
- Block number.
- Subdivision or project name.
- Lot area.
- Lot status.
- Relevant reservation information, where applicable.

- Date of status updates, where recorded.
- User responsible for a modification, where supported by the system.

The researchers examined how lot information was organized and how updates were reflected in existing records. Differences in the timing or recording of updates highlighted the importance of maintaining consistent lot information.

The findings from document analysis contributed to the design of the database schema and the identification of information fields needed by the system. The selected fields were aligned with the organization's monitoring requirements and the functions implemented in the application.

3.5.4 Requirements Gathering and Validation

After collecting information through interviews, observation, and document analysis, the researchers consolidated the findings to identify the requirements of the Web-Based Lot Availability Monitoring System. Requirements validation was intended to ensure that the identified functions reflected the operational needs of Golden Dragon Estate Corporation and were consistent with the system's scope.

The requirements were classified into functional and non-functional requirements.

Functional Requirements

Functional requirements describe the operations that the system is expected to perform. Based on the identified needs and the functions included in the application, the requirements covered the following areas:

1. User authentication and login.
2. Role-based access control for authorized personnel.
3. Lot monitoring and visual status presentation.
4. Lot status updating by authorized users.
5. Search and filtering of lot information, if implemented.
6. Dashboard and summary reporting, if implemented.
7. User account management, if implemented.
8. Activity logging and record tracking, if implemented.
9. Email notifications, if implemented.
10. Storage and retrieval of lot and user information.

These requirements provided a basis for determining the system's functional scope and guided the development and testing of its components.

Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational expectations of the system. The identified requirements included:

1. **Security:** The system should restrict access to protected functions and information to authorized users.
2. **Reliability:** The system should perform its intended operations consistently under normal operating conditions.
3. **Usability:** The interface should be understandable and easy for intended users to navigate.
4. **Performance:** The system should respond within an acceptable time during common operations.
5. **Scalability:** The system design should allow for reasonable future expansion where supported by the architecture.
6. **Maintainability:** The application should be organized to support troubleshooting, updates, and improvements.
7. **Availability:** The system should be accessible to authorized users when the hosting environment and network are operational.
8. **Data Integrity:** Lot information should be stored and updated consistently through appropriate validation and database operations.

The requirements were reviewed against the information collected from the organization's workflow and records. The validated requirements then served as a reference for system design, implementation, testing, and evaluation.

3.5.5 Importance of Data Gathering in the Study

Data gathering provided the foundation for understanding the existing lot monitoring process and identifying the operational requirements of Golden Dragon Estate Corporation. Through interviews, observation, document analysis, and requirements validation, the researchers obtained information relevant to the design and development of the system.

The information gathered helped determine the system's main functions, database requirements, interface design, and access-control needs. It also helped ensure that the development process was guided by the organization's workflow rather than relying solely on assumptions made by the researchers.

Furthermore, comparing information from different data-gathering methods helped identify common concerns and clarify system requirements. This approach strengthened the basis for the system's design and supported the development of a digital alternative to the company's manual lot monitoring process.

3.6 Testing Procedures

Testing was conducted to determine whether the developed Web-Based Lot Availability Monitoring System performed according to its specified requirements. The testing process focused on identifying defects, verifying the behavior of individual functions, examining the interaction between system components, and determining whether the application supported the intended lot monitoring activities.

The testing procedures were organized into three main activities: unit testing, system testing, and User Acceptance Testing (UAT). These activities addressed different aspects of software quality, from individual components to the integrated application and its suitability for intended users.

3.6.1 Unit Testing

Unit testing focused on examining individual functions or modules to determine whether they performed their intended operations. The researchers checked the behavior of relevant components during development before or as they were integrated into the complete application.

The following modules and functions were considered for unit testing, where applicable.

Authentication Module

The authentication functions were checked to determine whether login and logout operations behaved as expected.

Test scenarios included:

- Entering valid login credentials.
- Entering an incorrect username or account identifier.
- Entering an incorrect password.
- Logging out of an authenticated session.
- Checking session behavior after authentication.

The expected behavior was successful authentication for valid credentials and appropriate rejection or error handling for invalid credentials. Session-related checks depended on the authentication mechanism implemented in the application.

Lot Monitoring Module

The lot monitoring functions were checked to determine whether the application displayed lot information and status indicators correctly.

Test scenarios included:

- Displaying available lots.
- Displaying pending lots.
- Displaying sold lots.
- Retrieving lot information from the database.
- Displaying the correct visual indicator for each recorded status.

The expected behavior was that the interface would display the lot information and corresponding status based on the data returned by the backend.

Lot Status Management Module

The lot status management functions were checked to determine whether authorized users could submit valid status changes and whether those changes were stored correctly.

Test scenarios included:

- Changing a lot's status from available to pending.
- Changing a lot's status from pending to sold.
- Changing a sold lot back to available, if the application's rules permitted the operation.
- Verifying that valid status changes were saved.
- Checking how the interface responded to invalid or unauthorized requests.

The expected behavior was that permitted changes would be processed and stored, while invalid or unauthorized operations would be handled appropriately. The interface would display the resulting status after receiving the server's response.

User Management Module

If user management was implemented, its functions were checked to determine whether authorized administrators could perform the available account-management operations.

Test scenarios included adding users, editing account information, changing assigned roles, and deactivating accounts, where these functions were supported by the system.

The expected behavior was that permitted account operations would be processed correctly and that access restrictions would be maintained.

Email Notification Module

If email notifications were implemented, the notification functions were checked using the configured email service.

Test scenarios could include triggering supported account-related messages, password recovery messages, or other configured system notifications. The results were checked to determine whether the application generated the expected message and whether delivery to the intended recipient could be verified.

Only notification functions actually implemented in the application should be listed in the final testing documentation.

Dashboard Module

If a dashboard was implemented, its displayed summaries were checked against the corresponding lot records.

Test scenarios included:

- Displaying the total number of available lots.
- Displaying the total number of pending lots.
- Displaying the total number of sold lots.
- Checking whether dashboard summaries reflected changes to the underlying records.

The expected behavior was that the displayed totals would correspond to the applicable records and update according to the application's data-refresh mechanism.

3.6.2 System Testing

System testing examined the integrated application to determine whether its components worked together according to the identified requirements. It focused on the behavior of the frontend, backend, database, and other implemented services when used as a complete system.

The system testing activities included the following areas.

Functional Testing

Functional testing examined whether the implemented features performed their intended operations. Depending on the system's scope, the tests covered authentication, lot monitoring, lot status updates, user management, dashboard reporting, and email notifications.

Each function was checked using appropriate input scenarios to determine whether the application produced the expected response.

Integration Testing

Integration testing examined communication between connected system components. The researchers checked relevant interactions involving:

- Frontend requests and backend responses.
- API endpoints and application logic.
- Backend operations and database connectivity.
- Email service integration, where implemented.

The purpose was to determine whether submitted information was processed correctly and whether database operations and resulting interface responses behaved as expected.

Performance Testing

Performance testing examined the system's responsiveness during common operations. Relevant measures could include page-loading time, database query duration, response time for lot-related actions, and behavior under simultaneous access.

Performance results should be reported using actual measurements or documented observations. Claims about high performance or acceptable response times should be supported by test conditions, measurements, or clearly defined evaluation criteria.

Security Testing

Security testing examined whether the implemented authentication and access-control mechanisms behaved as intended.

Checks could include:

- Handling of invalid login attempts.
- Access to protected pages and functions.
- Enforcement of role-based permissions.
- Session handling and logout behavior.
- Protection against unauthorized modification of lot records.

These checks helped identify weaknesses in access restrictions and application behavior. The conclusions drawn from security testing should be limited to the controls and scenarios actually examined.

Compatibility Testing

If compatibility testing was performed, the application was checked in the browsers and devices included in the test plan. The purpose was to identify interface or functional issues that might occur in different supported environments.

The final documentation should name only the browsers and devices that were actually tested. Google Chrome, Microsoft Edge, and Mozilla Firefox should not be reported as tested environments unless the researchers verified the application in each browser.

3.6.3 User Acceptance Testing (UAT)

User Acceptance Testing was used to determine whether the system's implemented functions supported the tasks and requirements of its intended users. Unlike unit testing and system testing, which focus on technical behavior, UAT emphasizes whether users can complete relevant tasks using the application.

For this study, the UAT process involved selected personnel from Golden Dragon Estate Corporation, provided that their participation was part of the actual evaluation procedure. Participants were asked to perform relevant system tasks and assess whether the functions met their operational needs.

The evaluation activities could cover the following areas:

Ease of Use

Participants assessed whether the interface was understandable and whether navigation and common tasks could be performed without unnecessary difficulty. Relevant indicators included menu clarity, navigation, and ease of accessing lot information.

Functionality

Participants examined whether the available features performed their intended tasks, including viewing lot information, updating statuses where authorized, and accessing supported summaries or reports.

Efficiency

Participants considered whether the system made common lot monitoring tasks easier to perform. Relevant indicators included the effort required to locate lot information, update records, and retrieve data compared with the existing manual process.

If the study claims that the system reduced task completion time, this should be supported by measured task times or other documented comparative evidence.

User Satisfaction

Participants provided feedback on their overall experience, perceived usefulness, and willingness to use the system for relevant work activities.

Feedback collected during UAT was reviewed to identify issues, usability concerns, and opportunities for improvement. Any changes made in response to this feedback were checked as appropriate.

The UAT findings provided evidence about the system's acceptance among the participating users. The scope of the conclusions depended on the number and characteristics of participants, the tasks performed, and the evaluation method used.

3.6.4 Bug Identification and Resolution

During testing, identified software defects were documented and examined to determine their causes. The researchers followed a corrective process intended to resolve defects and verify the behavior of the affected functions.

The process involved the following steps:

1. Identifying and recording the issue.
2. Reproducing the issue under the relevant conditions.
3. Investigating the probable cause.
4. Implementing a correction.
5. Retesting the affected functionality.
6. Checking whether the issue was resolved and whether the correction introduced other problems.

This process supported iterative improvement throughout development and testing. Defects that remained unresolved were to be documented and considered when determining the system's readiness for use.

3.6.5 Summary of Testing Procedures

The testing procedures provided a structured approach to checking the developed Web-Based Lot Availability Monitoring System. Unit testing focused on individual functions, system testing examined the integrated application, and User Acceptance Testing assessed whether the system supported the needs of its intended users.

Together, these activities were intended to identify defects, verify expected behavior, and provide evidence about the application's suitability for lot availability monitoring. Conclusions about the system's reliability, security, performance, and acceptance should be based on the documented results of the tests actually performed

3.7 Evaluation Criteria

The evaluation criteria were established to assess the quality and suitability of the developed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. The evaluation focused on characteristics relevant to the application's purpose, including the accuracy of lot information, usability of the interface, consistency of operations, responsiveness, and protection of system data.

The evaluation approach should be aligned with the instruments and procedures actually used by the researchers. For this study, functionality, usability, reliability, efficiency, and security were identified as the principal areas of evaluation. Each area requires appropriate indicators and evidence to support the resulting assessment.

3.7.1 Functionality

Functionality refers to whether the system's implemented features perform the tasks for which they were designed. It was assessed by examining whether the application's operations corresponded to the defined functional requirements.

For the lot availability monitoring system, relevant indicators included:

1. The system displays lot information correctly.
2. Lot status changes are processed according to the applicable rules and user permissions.
3. Displayed status indicators correspond to the recorded lot statuses.
4. Implemented reports and dashboard summaries present accurate information.
5. Implemented user-management functions behave as intended.
6. The application handles invalid inputs and unsuccessful operations appropriately.

Functional quality was assessed through relevant test cases and documented observations. A favorable assessment would indicate that the tested functions performed according to their specified requirements

3.7.2 Usability

Usability refers to the extent to which intended users can understand and operate the system to complete relevant tasks. It is particularly important for a lot monitoring application because personnel must be able to locate and interpret lot information efficiently.

Relevant usability indicators included:

1. The interface is easy to understand.
2. Navigation between available functions is clear.
3. Lot information is presented in an understandable manner.
4. Color-coded indicators help users distinguish lot statuses.
5. Users can complete common tasks without unnecessary difficulty.

Usability was to be examined through the user-evaluation method selected for the study. If the researchers used the System Usability Scale (SUS), the standard ten-item questionnaire and its scoring procedure should be described in this section or in a separate subsection. If a researcher-developed questionnaire was used instead, its items, response scale, and interpretation method should be documented accurately.

A high usability assessment would suggest that the interface supports users in completing their intended tasks. The conclusion should be based on the actual evaluation responses rather than assumptions about the interface.

3.7.3 Reliability

Reliability refers to the system's ability to perform its intended functions consistently under specified operating conditions. It is important because personnel rely on accurate lot information when monitoring availability and responding to inquiries.

The evaluation considered indicators such as:

1. The system consistently retrieves lot information.
2. Valid lot status updates are saved correctly.
3. Information remains consistent between the database and the interface after successful operations.
4. The application handles common errors appropriately.
5. The system operates without frequent unexpected failures during the observed test conditions.

Reliability was examined through relevant functional and system tests and, where available, documented user feedback. Any claims about failure rates or long-term operational stability should be supported by appropriate observations or measurements.

3.7.4 Efficiency

Efficiency refers to how effectively the system performs its tasks in relation to the time and resources required. In the context of this study, efficiency concerns the effort needed to retrieve lot information, update statuses, and access relevant summaries.

Relevant indicators included:

1. The system responds within an acceptable time during common operations.
2. Lot information can be retrieved without unnecessary steps.
3. Authorized users can update lot statuses through the available interface.
4. The system reduces reliance on manually searching through physical records.
5. Common monitoring tasks can be completed efficiently.

Efficiency can be evaluated through measured response times, task completion times, observed task steps, or user assessments, depending on the actual research procedure. If the researchers intend to claim that the system is faster than the manual process, the comparison should use documented measurements or a clearly described evaluation method.

3.7.5 Security

Security refers to the protection of system information and functions against unauthorized access, modification, or disclosure. Because the system manages business-related lot information and user accounts, access controls and appropriate data-handling practices are important.

Relevant security indicators included:

1. Authentication operates as intended.
2. Protected functions are restricted to authorized users.
3. Role permissions are enforced for relevant operations.
4. Unauthorized requests to modify lot information are rejected.
5. User and lot records are handled through appropriate application and database controls.

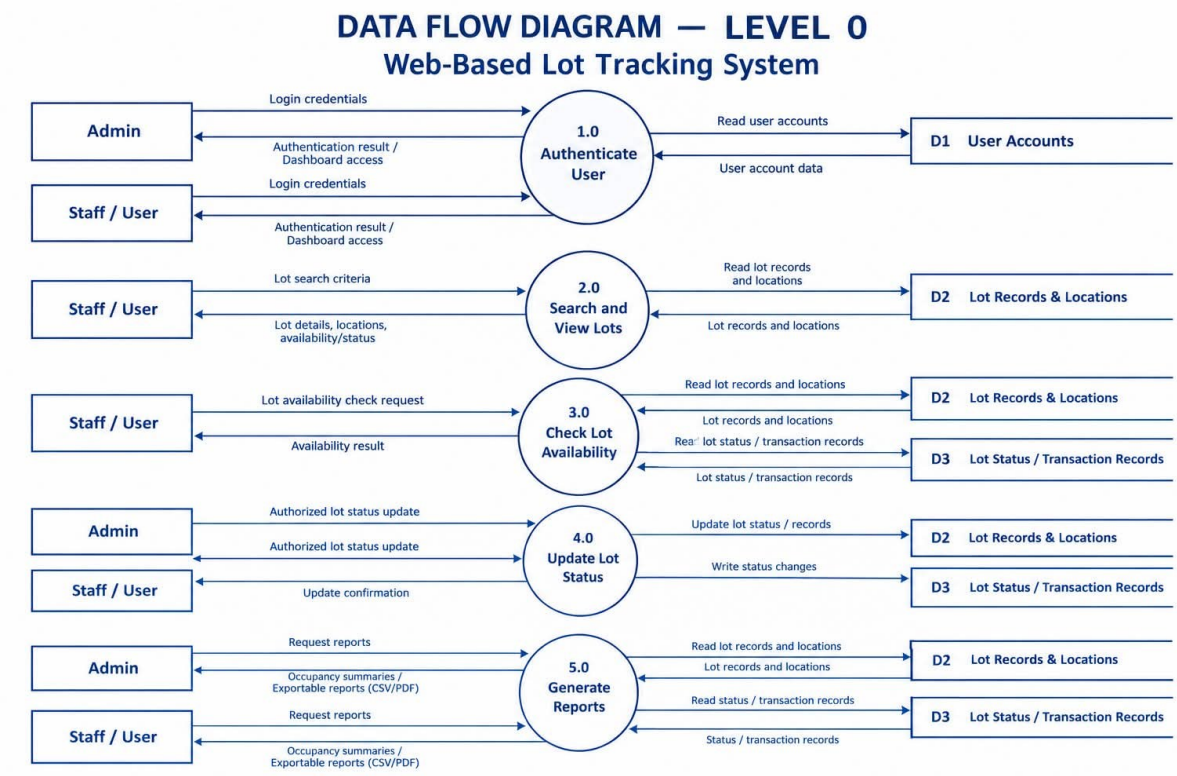
Security was assessed through the implemented controls and the relevant tests performed by the researchers. Claims regarding password protection, session management, or role-based access control should correspond to the actual implementation. A favorable result from selected tests does not establish that the system is completely secure; conclusions should be limited to the controls examined.

3.7.6 Summary of Evaluation Criteria

The evaluation criteria provided a framework for examining the developed system's functional correctness, usability, reliability, efficiency, and security. Each criterion addressed an aspect of quality relevant to the system's purpose of monitoring residential lot availability through a centralized web-based platform.

The evaluation findings were intended to identify areas in which the system met its requirements and areas that required improvement. When supported by documented test results and user feedback, the assessment could provide a basis for determining the system's suitability for the operational needs of Golden Dragon Estate Corporation.

Data Flow Diagram (DFD) Level 0



The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram, illustrates the overall interaction between the Web-Based Lot Availability Monitoring System and its external entities. It presents the system as a single process and identifies the primary users, the information they provide, and the information they receive during system operation.

Administrator

The Administrator is responsible for managing the system and performing administrative functions according to the assigned access permissions.

- The Administrator submits login credentials to access the system.
- The Administrator manages property and lot information, as well as user accounts, where these functions are supported by the system.
- The Administrator requests reports or views dashboard information, if these features are implemented.
- In response, the system provides login results, dashboard information, generated reports, and confirmation messages for successfully processed operations, as applicable.

Employee

The Employee accesses the system to perform authorized lot-monitoring tasks according to the permissions assigned to their account.

- The Employee submits login credentials to access the system.
- The Employee views lot information and performs permitted lot-status updates.
- The Employee enters or updates related records when the corresponding functions are available.
- In response, the system provides login results, current lot information, operation confirmations, and other information permitted by the employee's assigned role.

Web-Based Lot Availability Monitoring System

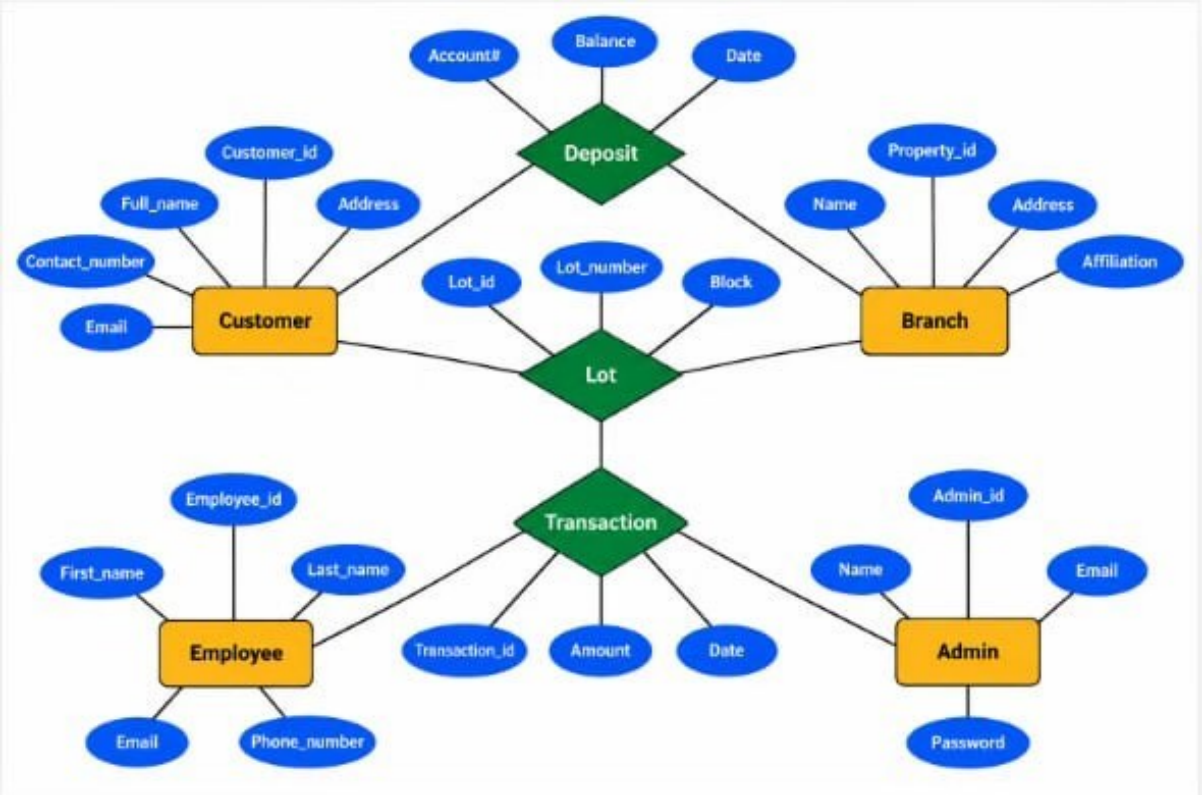
The Web-Based Lot Availability Monitoring System serves as the central application for processing and managing lot availability information. It receives user requests, validates submitted information, processes authorized operations, and communicates with the database to retrieve or update records.

The system presents lot information through a visual interface using color-coded status indicators: green for available lots, yellow for pending lots, and red for sold lots. It also returns appropriate responses to users after processing their requests. The accuracy of the displayed information depends on the validity of the stored records and the timely updating of lot statuses by authorized users.

Summary

The Level 0 DFD provides a high-level overview of the information exchanged between the Administrator, Employee, and the Web-Based Lot Availability Monitoring System. It illustrates how users access the application, submit requests, and receive the information needed for their assigned tasks. By centralizing lot availability records in a web-based platform, the system supports more organized information management and helps personnel monitor lot statuses at Golden Dragon Estate Corporation.

Entity Relationship Diagram (ERD)



The Entity Relationship Diagram (ERD) illustrates the logical structure of the database used by the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. It identifies the major entities, their attributes, primary keys, foreign keys, and relationships. The ERD serves as a guide for organizing system data and defining how related records are stored, accessed, and maintained.

Customers

The Customers entity stores information about individuals who inquire about or express interest in residential lots, provided that customer management is included in the system's implemented functions.

- It may contain customer information such as full name, contact number, email address, and residential address.
- Each customer is identified by a unique Customer ID, which serves as the primary key.
- If customer records are linked to reservations or purchases, the relationship should be established through the appropriate transaction or reservation records.

Lots

The Lots entity is a central component of the database because it stores information about individual residential lots monitored by the system.

- It contains attributes such as Lot ID, lot number, block number, lot area, and current availability status, according to the implemented database design.
- Each lot is identified by a unique Lot ID, which serves as the primary key.
- A Property ID may serve as a foreign key to associate each lot with its corresponding property or subdivision.
- Additional attributes, such as map coordinates or status-update dates, should be included only if they exist in the actual database.

The Lots entity supports the storage and retrieval of lot information needed to display and monitor availability through the system's interface.

Properties

The Properties entity stores information about the subdivisions or properties managed by Golden Dragon Estate Corporation, if this entity is included in the database.

- It may contain attributes such as Property ID, property name, location, and active status.
- Each property is identified by a unique Property ID, which serves as the primary key.
- The Property ID may be used as a foreign key in the Lots entity to associate individual lots with their corresponding property.

One property can contain multiple lots, establishing a one-to-many relationship between Properties and Lots. Each lot belongs to one property when the database is designed according to this relationship.

Transactions

The Transactions entity stores records of lot-related business activities, such as reservations or purchases, if transaction management is part of the implemented system.

- Each transaction is identified by a unique transaction identifier, which serves as the primary key.
- It may contain information such as transaction date, transaction type, payment method, and notes.
- Customer ID and Lot ID may serve as foreign keys to connect each transaction to the corresponding customer and lot.

This entity can support the recording and retrieval of transaction history. However, the relationship between transactions, customers, and lots should reflect the company's actual business process and the database structure.

Employees

The Employees entity stores information about employees who are authorized to access the system, if employee accounts are maintained in a separate table.

- It may contain attributes such as Employee ID, employee name, email address, and other required account information.
- Employee ID serves as the primary key.
- Login credentials and account-recovery fields should be represented according to the actual authentication implementation.

Sensitive authentication information, particularly passwords, should not be stored as plain text. Passwords should be handled using an appropriate password-hashing mechanism.

Administrators

The Administrators entity stores account information for users assigned administrative responsibilities, if administrators are maintained separately from employee accounts.

- Each administrator is identified by a unique Admin ID, which serves as the primary key.
- The entity may contain the account fields required for authentication and administrative access.
- Administrative responsibilities may include managing lot information, properties, and user accounts, depending on the functions implemented in the system.

If administrators and employees use a shared user table with different roles, they should not be represented as separate entities unless the actual database design supports that arrangement.

Relationships Among Entities

The relationships among the entities define how related records are connected within the database.

- **Properties and Lots:** One property can contain many lots, while each lot belongs to one property when this relationship is implemented using a Property ID foreign key.

- **Customers and Transactions:** One customer may have multiple transactions, while each transaction refers to the appropriate customer, if customer transaction management is implemented.
- **Lots and Transactions:** One lot may be associated with multiple transaction records over time, depending on the business rules. Each transaction should identify the lot involved.
- **Employees and Administrators:** These entities represent authorized system users only if the database maintains them as separate tables. If user roles are stored in a shared account table, the ERD should reflect that structure instead.

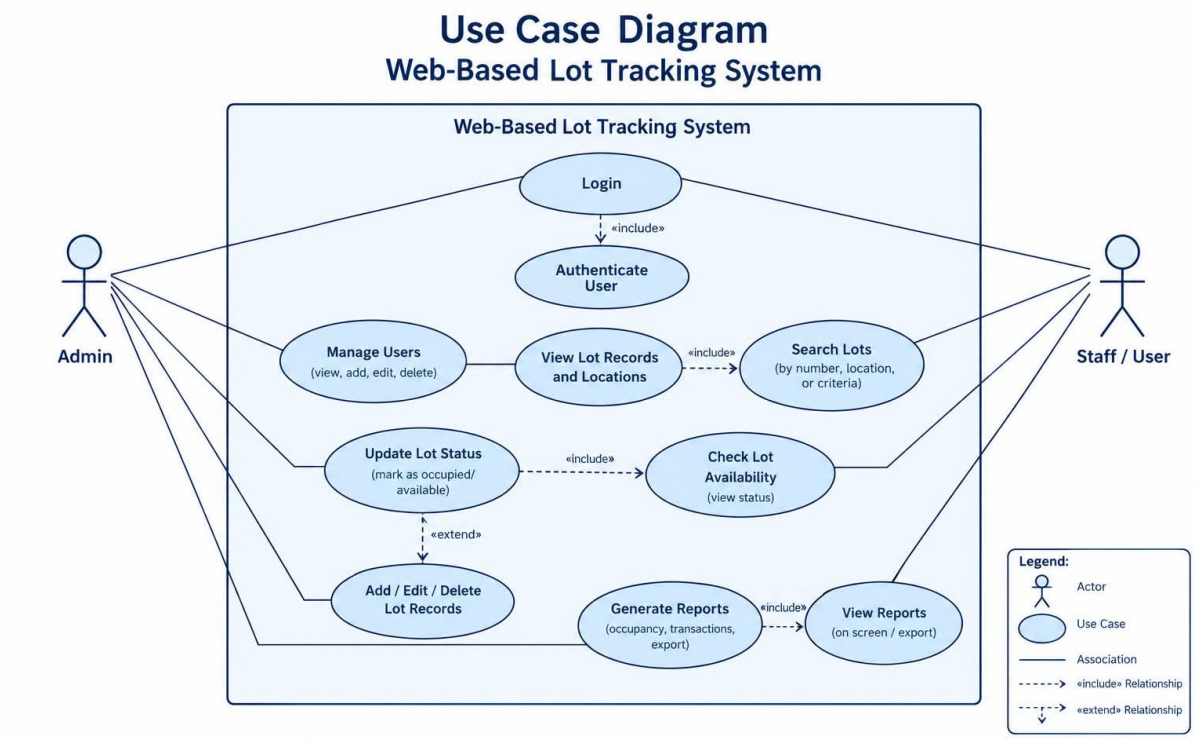
Primary keys provide unique identification for records, while foreign keys establish relationships between related tables. Appropriate constraints and database validation help maintain referential integrity and reduce inconsistent or duplicated records.

Summary

The Entity Relationship Diagram presents the logical database design of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. It illustrates how the system's data entities are organized and how their relationships support the storage and retrieval of lot availability information.

Through appropriately defined primary keys, foreign keys, and relationships, the database design helps maintain data consistency and supports the system's monitoring functions. The final ERD should accurately represent the entities and relationships implemented in the actual MySQL database.

Use Case Diagram



The Use Case Diagram illustrates the interactions between the users and the functions of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. It identifies the two primary actors, the Administrator and the Employee, and presents the system functions available to each actor according to their assigned roles and permissions.

The diagram provides an overview of how users access the application and perform their authorized tasks. It also illustrates the role of authentication and access control in limiting access to protected functions.

Administrator

The Administrator is responsible for managing the system's administrative functions and maintaining the accuracy of lot availability information.

- Login: Authenticates the administrator before granting access to protected system functions.
- Manage Lots: Adds, edits, or updates lot information and availability statuses.
- Manage Properties: Maintains property or subdivision information, if supported by the system.
- Manage Employee Accounts: Creates, updates, or deactivates employee accounts, if implemented.
- View Dashboard: Accesses available summaries of lot statuses and other system information.

- View Records: Accesses relevant records according to the administrator's permissions and the implemented features.
- Generate Reports: Produces system reports if reporting functionality is available.
- Logout: Ends the administrator's session after completing system operations.

Employee

The Employee uses the system to perform assigned lot-monitoring tasks according to the permissions granted to their account.

- Login: Authenticates the employee before accessing authorized system functions.
- View Lot Information: Views lot details and availability statuses through the system interface.
- Update Lot Status: Changes lot statuses when the employee has permission to perform the operation.
- View Dashboard: Accesses the dashboard information available to the employee's role.
- Manage Related Records: Performs customer or transaction-related operations only if these functions are included in the implemented system.
- Generate Reports: Accesses or generates reports if the employee's role permits it and the feature is available.
- Logout: Ends the employee's session after completing assigned tasks.

Common System Functions

The following functions may be shared by the Administrator and Employee, depending on the system's access-control configuration.

- Login: Verifies user credentials before granting access to protected functions.
- View Dashboard: Displays available summaries and relevant lot-monitoring information.
- View Lot Information: Presents lot details and their current recorded statuses.
- Logout: Ends the user's session to help prevent unauthorized access to protected functions.

Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is an access-control approach in which permissions are assigned according to a user's role. In the Web-Based Lot Availability Monitoring System, the Administrator and Employee roles determine which protected functions each user is authorized to access.

The Administrator may be permitted to manage system settings, user accounts, properties, and lot information, depending on the implemented functions. Employees are granted access to the functions necessary for their assigned responsibilities, such as viewing lot information and updating lot statuses where authorized.

By enforcing role-based permissions, the system can help restrict access to administrative functions and reduce the risk of unauthorized changes to lot records. The effectiveness of this mechanism depends on the actual implementation of authentication and authorization checks.

Summary

The Use Case Diagram presents the functional interactions between the Administrator, Employee, and Web-Based Lot Availability Monitoring System. It identifies the actions available to each actor and illustrates how access to system functions is controlled according to assigned permissions.

The diagram supports the documentation of the system's functional requirements and clarifies the responsibilities of its users. Through authorized access to centralized lot information, the system aims to support organized lot monitoring and information management at Golden Dragon Estate Corporation.

CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the development, implementation, testing, evaluation, results, and discussion of the **Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation**. It describes the developed system and explains how it addresses the limitations of the company's previous manual process of monitoring residential lot availability using printed subdivision maps and handwritten status updates. The system was developed to provide a centralized platform for managing lot information and other related records, helping improve the efficiency and organization of the company's lot monitoring operations.

The development of the system followed the Agile Software Development Methodology discussed in Chapter 3. The researchers used an iterative approach to system planning, design, development, and testing to align the application with the identified requirements. The system was developed using web technologies and a relational database to support the storage, management, and retrieval of information. Its interface also provides a visual representation of lot availability through color-coded lot statuses, allowing authorized users to identify the current status of residential lots more easily.

This chapter discusses the system's implemented modules, including user authentication, dashboard, property management, lot management, customer records, transaction recording, report generation, and email notifications, where applicable. It explains the purpose and operation of each implemented module and how the modules work together to support the company's lot monitoring activities. The chapter also presents the hardware and software requirements, system architecture, and database design used in developing and operating the application.

Furthermore, the chapter presents the testing procedures and evaluation results used to assess the system's functionality, usability, reliability, and other relevant quality characteristics. The findings are discussed based on the actual testing activities and evaluation data gathered by the researchers. These results provide a basis for determining whether the developed system meets the identified requirements and supports the intended users' operational needs.

Finally, this chapter discusses the results of the system implementation in relation to the objectives of the study. It highlights the improvements provided by the application over the previous manual process, particularly in organizing lot information, monitoring availability, retrieving records, and managing related information. The discussion also identifies the system's limitations and areas for improvement based on the actual implementation and evaluation findings.

4.2 Overview of the Developed System

The Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation was developed to improve the company's existing process of monitoring residential lot availability. The previous process relied on printed subdivision maps and handwritten status updates, which could make it difficult to maintain consistent and up-to-date lot information. The developed system provides a centralized web-based platform where authorized users can view, manage, and update lot records through a digital interface.

The primary purpose of the system is to support the company's daily lot monitoring and information management activities. It organizes lot-related information in a centralized database, allowing authorized users to retrieve and update records without depending entirely on physical documents. This approach helps minimize the risks of duplicate records, inconsistent status updates, misplaced documents, and manual encoding errors.

The system follows a client-server architecture. The client-side interface was developed using React and Tailwind CSS, while the server-side operations were implemented using Node.js and Express.js. MySQL serves as the relational database management system for storing and retrieving application data. This architecture separates the user interface, application logic, and data storage, allowing the system's components to work together in processing user requests and maintaining records. Nodemailer with SMTP is also used for email notifications for selected system events, where this functionality is implemented and operational.

To manage access to the system, Role-Based Access Control (RBAC) is used to assign permissions according to user roles. Administrators have access to administrative functions, such as managing employee accounts and maintaining property and lot records, depending on the permissions implemented in the application. Employees are given access to the operational functions required for their assigned responsibilities. These permissions help organize user access and limit the availability of functions according to each user's role.

The dashboard provides an overview of important lot availability information, including the numbers of available, pending, and sold residential lots, if these indicators are enabled in the deployed application. The system also uses color-coded indicators to help users distinguish lot statuses visually. Green represents available lots, yellow represents pending lots, and red represents sold lots. This visual presentation allows authorized users to identify lot availability more easily than by examining handwritten updates on printed maps.

The lot monitoring module is one of the system's primary components. It allows authorized users to view lot information and update lot statuses through the web interface. When a user submits a status change, the application processes the request and updates the corresponding database record. The displayed information can then reflect the updated status based on the application's refresh and data-retrieval behavior. This process provides a digital alternative to manually changing lot indicators on printed subdivision maps.

The property management module organizes information about the real estate properties managed by the company. Where supported by the implemented system, authorized users can add, view, and update property records. Each property can be associated with its corresponding residential lots, enabling the company to organize lot information according to its properties or subdivisions.

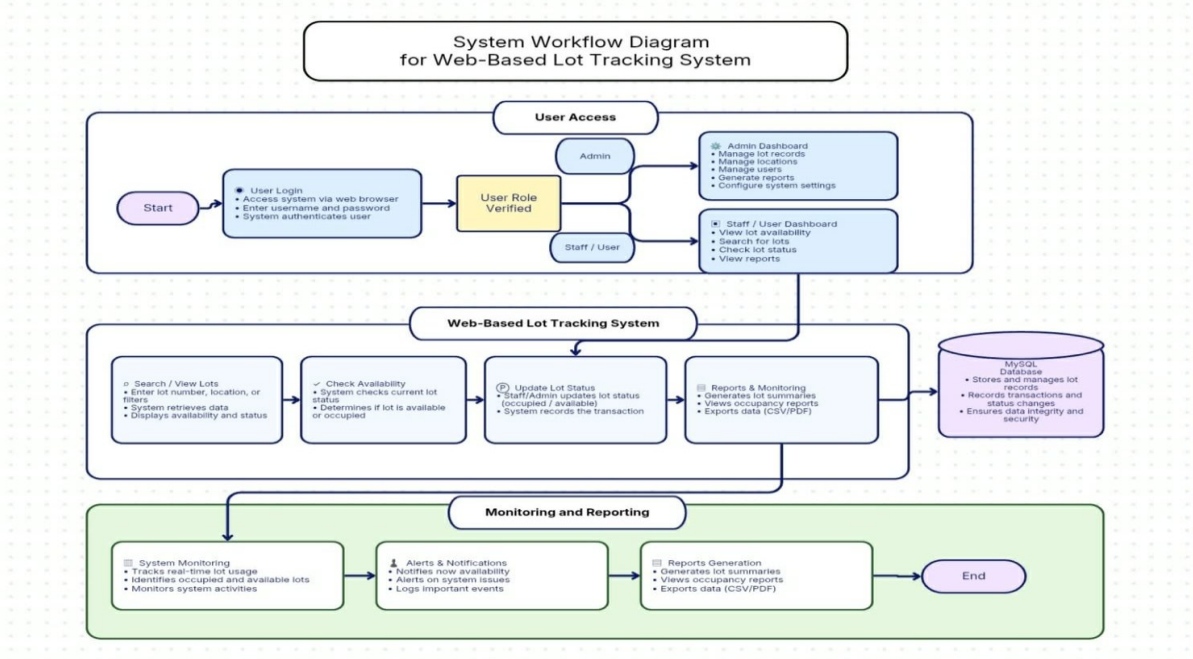
The customer management module provides a centralized location for maintaining customer records associated with the company's lot-related activities. Authorized users can access and maintain customer information through the application, subject to their assigned permissions. Centralizing these records helps make customer information easier to retrieve and manage during daily operations.

The transaction management module supports the recording and retrieval of lot-related transactions, provided that these functions are available in the deployed application. Transaction records can be associated with the relevant customer and residential lot to maintain an organized history of recorded activities. This arrangement helps authorized personnel review transaction information when needed and supports the systematic management of business records.

The report generation module, if implemented and operational, allows authorized users to generate summaries from stored system records. Depending on the available report functions, these summaries may include lot availability, customer records, and transaction information. The resulting reports can assist personnel in reviewing operational information and support management decision-making.

The system also incorporates an email notification function using Nodemailer and SMTP for selected events, where configured. This feature supports the delivery of relevant updates through email and can reduce the need for separate manual communication for events covered by the notification function.

Overall, the developed Web-Based Lot Availability Monitoring System integrates lot monitoring, user access management, and related information management functions into a centralized web application. Its implementation provides Golden Dragon Estate Corporation with a digital means of maintaining lot information and monitoring residential lot availability. The system's effectiveness in meeting its intended requirements is discussed further through the testing procedures and evaluation results presented in the succeeding sections.



4.3 System Development Methodology

The development of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation followed the Agile Software Development Methodology, which emphasizes iterative development, continuous improvement, and the gradual implementation of system functionalities. This methodology allowed the researchers to organize the development process into manageable phases, from identifying the company's operational requirements to designing and implementing the system. It also provided opportunities to review completed components and make necessary revisions during development.

The application of the Agile methodology in this study involved the following phases:

4.3.1 Requirements Gathering

The Requirements Gathering phase focused on identifying the operational needs of Golden Dragon Estate Corporation and translating them into system requirements. During this phase, the researchers gathered information about the company's existing lot monitoring procedures through the data-gathering methods described in Chapter 3. These activities helped the researchers understand how the company monitored residential lot availability using printed subdivision maps and manually updated lot statuses.

The existing process presented challenges associated with maintaining accurate and consistent records, updating lot information, and organizing physical documents. These observations helped the researchers identify areas where a computerized monitoring system could support the company's daily operations.

Based on the information gathered, the researchers identified the functional requirements of the system, including user authentication, lot availability monitoring, and the management of related records. Other functions, such as property management, customer record management, transaction recording, report generation, and email notifications, were included where they formed part of the verified system requirements and implemented application. Non-functional requirements, including security, usability, reliability, performance efficiency, and data integrity, were also considered during development.

The information gathered during this phase served as the basis for designing the database, user interface, and application modules. By examining the company's existing workflow, the researchers established the system requirements intended to address the identified operational needs.

4.3.2 System Planning

The System Planning phase established the project's scope, objectives, development priorities, and resource requirements. The researchers organized the development activities into manageable tasks based on the system requirements identified during the preceding phase. This approach helped structure the implementation of the application's major components and establish an organized development workflow.

The development priorities included essential functions such as user authentication and lot management, together with other required modules for handling property information and related operational records. The order of implementation was determined by the requirements and dependencies among the system components.

The researchers also selected the software tools and technologies used to develop the application. React was used to build the client-side interface, while Tailwind CSS supported the styling and layout of the user interface. Node.js and Express.js were used to implement the server-side logic and application programming interfaces (APIs). MySQL served as the relational database management system for storing and retrieving application records. Nodemailer with SMTP was used for email notifications where this functionality was implemented.

The planning activities provided a framework for organizing development tasks and allocating available resources. As development progressed, the researchers reviewed the implemented components and made necessary adjustments to address identified issues and maintain consistency with the system requirements.

4.3.3 System Design

The System Design phase translated the identified system requirements into models and interface layouts that guided application development. During this phase, the researchers prepared design models to represent the system's data flow, database structure, and interactions between users and system functionalities.

The Data Flow Diagram (DFD) illustrated the movement of information between the system and its external entities. The Entity Relationship Diagram (ERD) represented the logical organization of the database, including its entities, attributes, primary keys, foreign keys, and relationships. The Use Case Diagram identified the interactions between the system's user roles and the functions available to them according to their assigned permissions.

The researchers also designed the user interface to provide a consistent and understandable navigation structure. The interface was organized to help authorized users access relevant functions and manage lot information through the web application. Color-coded indicators were incorporated into the lot monitoring interface to distinguish the different lot statuses: green for available lots, yellow for pending lots, and red for sold lots.

The resulting design models and interface layouts served as references during implementation. They helped guide the organization of application components, database records, and user interactions while maintaining alignment with the identified requirements and objectives of the study.

Software Development

The Software Development phase involved implementing the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation based on the requirements and design models established during the earlier phases. Following the Agile Software Development Methodology, the researchers organized the implementation into functional modules. These modules were developed and reviewed progressively to identify issues and make necessary revisions throughout the development process.

The client-side interface was developed using React, which supports the creation of reusable user interface components. It was used to organize the application's pages and interactive elements, allowing users to access lot information and other available system functions through a web interface.

Tailwind CSS was used to style the application's interface. Its utility-based styling approach helped maintain consistency in the layout, spacing, typography, and appearance of interface components. The resulting interface was designed to help authorized users navigate the application and perform their assigned tasks.

The server-side application was implemented using Node.js and Express.js. These technologies handled server-side operations and supported communication between the frontend and the database through application programming interfaces (APIs). The backend processed requests associated with the implemented system functions, including authentication, lot management, and the management of other records supported by the application.

MySQL was used as the relational database management system for storing and retrieving application data. The database structure was designed to organize the records required by the system and establish relationships between related data. Primary keys, foreign keys, and appropriate database constraints were used where applicable to support data consistency and maintain the relationships among records.

Nodemailer with SMTP was incorporated for email notifications for selected system events, provided that the notification functionality was configured and operational in the deployed application. This feature supports the delivery of system-related messages through email for events covered by the implemented notification process.

Throughout development, the researchers reviewed the implemented components and addressed identified issues as needed. The development process focused on aligning the application's functionality with the identified requirements and integrating its components into a working web-based lot availability monitoring system.

Testing

The Testing phase was intended to verify whether the developed Web-Based Lot Availability Monitoring System performed according to its specified requirements. Testing activities were used to identify defects, check the behavior of implemented functions, and examine the interaction between the application's components. The testing methods described below should reflect the procedures actually performed by the researchers.

Unit Testing

Unit testing involved checking individual functions or components independently. Depending on the tests conducted, these checks could include verifying user authentication, lot status updates, record management, and

other implemented functions. The purpose was to identify errors within individual components before or during their integration with the rest of the application.

Integration Testing

Integration testing examined the interaction among the frontend, backend, and MySQL database. The process involved checking whether requests submitted through the user interface were processed by the backend and whether the resulting data were stored or retrieved correctly. These checks helped determine whether the connected components worked together as intended.

System Testing

System testing evaluated the application as an integrated system. The researchers checked the availability and behavior of implemented features under the conditions covered by the test procedures. These checks could include authentication, dashboard information, lot monitoring, and other operational modules included in the deployed application. Any additional functions, such as report generation or email notifications, should be included only if they were implemented and tested.

User Acceptance Testing (UAT)

User Acceptance Testing was intended to determine whether the system met the operational needs of its intended users. Where UAT was conducted, representative users from Golden Dragon Estate Corporation interacted with the application and assessed its suitability for their tasks. Feedback and observations gathered during this process could be used to identify issues and areas requiring improvement.

The results of the testing activities should be presented in the relevant testing and evaluation sections of this chapter. These results should be based on recorded test cases, observed system behavior, and actual feedback rather than assumptions about the application's performance.

Deployment

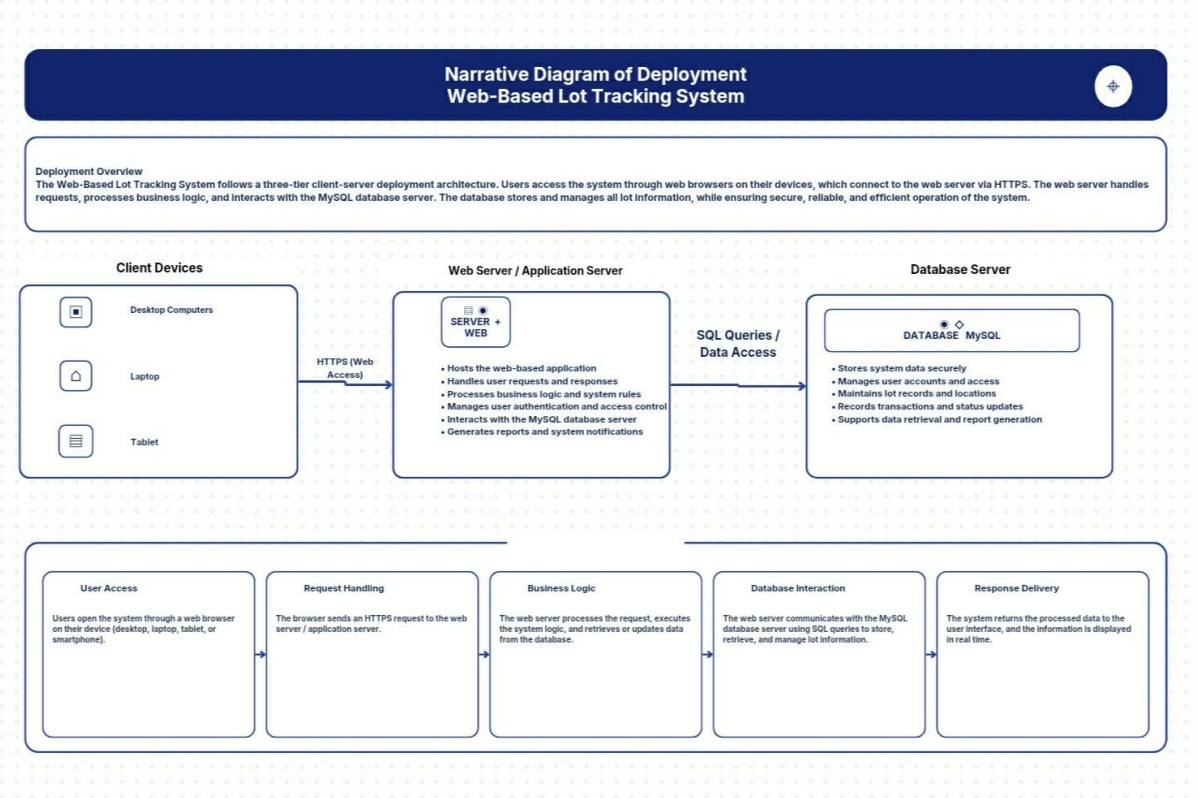
The Deployment phase involved making the Web-Based Lot Availability Monitoring System available for operational use at Golden Dragon Estate Corporation. The deployment process included preparing the application environment, configuring the required software components, and establishing the connection between the frontend, backend, and MySQL database.

The frontend was configured to communicate with the backend through the application's API endpoints. The backend was also configured to access the database to process the data required by the system. User accounts and access permissions were established according to the roles supported by the application to help control access to its functions.

After deployment, the system's availability and operation were checked within the configured environment. Any verification activities performed during this stage should be documented according to the actual checks conducted by the researchers. Features that were not implemented or verified during deployment should not be presented as successfully tested.

As a web-based application, the system can be accessed through a compatible web browser using the appropriate network connection and access address. This allows authorized personnel to use the application without installing a separate desktop program, provided that the deployed server and other required components are available.

The deployment made the system available for its intended lot monitoring activities. Its operational suitability and performance should be discussed in relation to the actual testing and evaluation findings presented in the succeeding sections.



Maintenance

The Maintenance phase focuses on preserving the functionality, security, and reliability of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation after deployment. Maintenance activities may include correcting software defects, addressing operational issues, updating software components, and improving existing functions to support the organization's requirements.

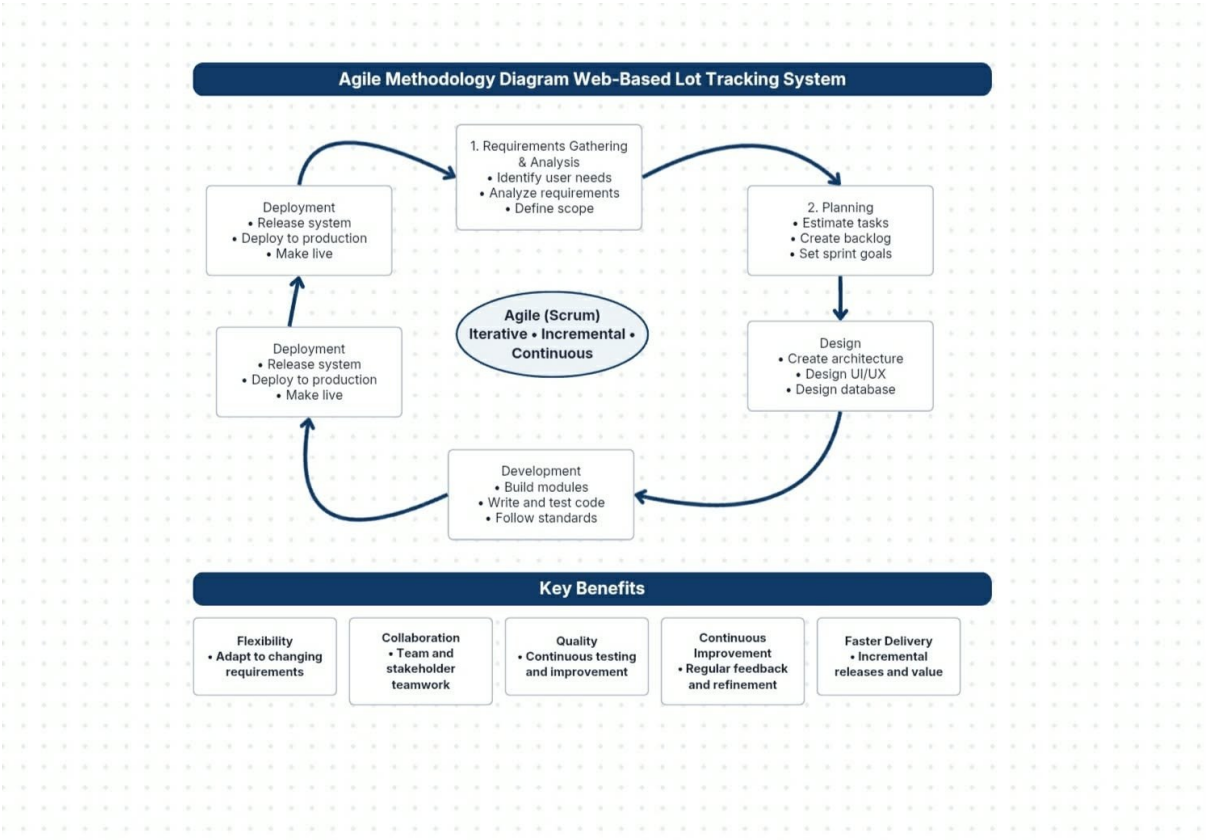
Corrective maintenance involves identifying and fixing software defects or unexpected system behavior discovered during operation. When an issue is reported or detected, the affected component can be examined and modified to restore its intended functionality. These corrections help maintain the proper operation of the system's implemented modules.

Preventive maintenance focuses on reducing the risk of future problems. Recommended activities include reviewing software dependencies, applying appropriate security updates, checking user access permissions, and verifying that database backup procedures are functioning correctly. Regular database backups and integrity checks can also help protect stored information against accidental loss or corruption. The frequency of these activities should be determined according to the organization's operational needs and available resources.

Adaptive maintenance involves modifying the system when changes in the operating environment or organizational requirements make adjustments necessary. Examples may include updating software dependencies, accommodating changes in the database environment, or modifying application components to maintain compatibility with supported technologies.

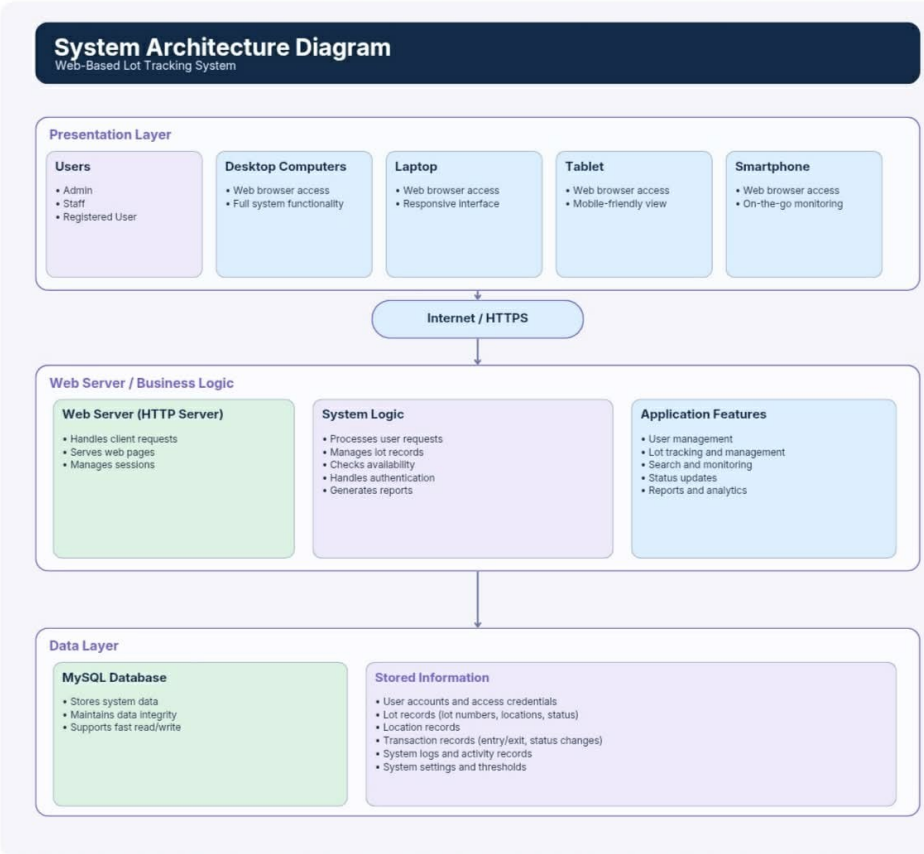
Perfective maintenance focuses on improving existing system functions and the overall user experience. Possible improvements include refining the interface, simplifying navigation, enhancing lot information displays, and adding functions that address newly identified operational needs. Any proposed enhancements should be evaluated and implemented according to their relevance, feasibility, and compatibility with the existing application.

The system's modular structure provides a foundation for future modifications by allowing individual components to be reviewed and updated as needed. Through appropriate maintenance planning and implementation, Golden Dragon Estate Corporation can continue to use the application for lot availability monitoring while addressing operational issues and adapting to future requirements.



4.4 System Architecture

The **Web-Based Lot Availability Monitoring System** adopts a **three-layer client-server architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural approach separates the user interface, application processing, and database management into independent but interconnected components. Such separation improves system organization, simplifies maintenance, enhances scalability, and allows each layer to perform its designated responsibilities without interfering with other components.



The Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer. Each layer has a distinct responsibility in processing user requests, managing application logic, and storing system information. This separation organizes the system components and supports communication between the user interface, server-side application, and database.

When a user performs an operation, the request is submitted through the presentation layer and processed by the application layer. The application layer validates the request, applies the relevant business rules, and communicates with the data layer when information needs to be retrieved or modified. The resulting information is then returned to the presentation layer for display. This process supports the organized management of lot availability information and related records.

Presentation Layer

The Presentation Layer serves as the user interface of the Web-Based Lot Availability Monitoring System. It presents information and receives user input through interactive web pages. The interface was developed using React and Tailwind CSS and is accessed through a compatible web browser.

The presentation layer contains the visual components used to interact with the system. Depending on the implemented functions, these components include the login page, dashboard, lot monitoring interface, and screens for managing property information and other related records. The interface is organized to help authorized users navigate the application and access the functions permitted by their assigned roles.

One of the main components of the presentation layer is the lot monitoring interface. Residential lots are displayed using color-coded indicators to represent their recorded availability status. Green indicates an available lot, yellow indicates a pending lot, and red indicates a sold lot. These visual indicators help users identify lot statuses without relying solely on printed subdivision maps or manually written updates.

The presentation layer also collects user input and may perform client-side validation, where implemented, before submitting requests to the server. The application layer remains responsible for validating requests and enforcing the relevant business rules before data are modified.

Overall, the presentation layer provides the interface through which users access the system's available functions and view lot availability information.

Application Layer

The Application Layer handles the server-side processing and business logic of the system. It receives requests from the presentation layer, processes the requested operations, applies validation rules, and communicates with the database when information must be retrieved or updated.

This layer was developed using Node.js and Express.js. Node.js provides the runtime environment for the server-side application, while Express.js supports the implementation of routes and APIs used for communication between the frontend and backend.

When a user performs an operation, such as logging in or updating a lot's status, the application layer processes the request according to the system's implemented logic. Authentication and access permissions are applied to restrict functions according to the user's assigned role, where these controls are implemented. The application layer also handles other supported operations, such as maintaining lot records and managing related information.

The application layer communicates with the MySQL database to retrieve existing records and save authorized changes. It processes the resulting information before returning an appropriate response to the user interface. This arrangement separates the application's processing logic from its visual components and database storage.

If the email notification feature is implemented and configured, the application layer also coordinates with Nodemailer and the SMTP service to send messages for the system events supported by the application.

By separating server-side processing from the presentation and data layers, the architecture provides an organized structure for developing, maintaining, and modifying the application.

Data Layer

The Data Layer manages the storage, organization, retrieval, and modification of information used by the Web-Based Lot Availability Monitoring System. MySQL serves as the relational database management system, storing the records required by the application's implemented functions.

The database contains the tables and relationships defined in the system's actual database design. These may include records for user accounts, properties, residential lots, customers, and transactions, depending on the entities implemented in the deployed database. Primary keys identify records, while foreign keys and other appropriate constraints establish relationships and help maintain referential integrity.

When a user performs an operation that requires data access, the application layer sends the appropriate database query. The database then retrieves or modifies the requested records and returns the result to the application layer for further processing. This approach centralizes data management and supports consistent access to stored information.

Access to the database is handled through the server-side application rather than through direct database connections from ordinary users' browsers. Appropriate database permissions and application-level access controls help protect organizational information against unauthorized access or modification.

Database backups, integrity checks, and other maintenance activities may also be used to protect stored information, depending on the procedures established by the organization. These activities should be described as existing practices only if they have actually been configured or performed.

Overall, the data layer provides the structured storage required to maintain lot availability information and other records managed by the application.

4.5 Hardware and Software Requirements

The successful development, deployment, and operation of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation require suitable hardware and software resources. These resources provide the computing environment needed to develop the application, manage its database, and allow authorized users to access its functions.

The hardware requirements describe the physical equipment needed to develop and operate the system, while the software requirements identify the operating systems, programming technologies, database software, development tools, and web browsers used in its implementation. The actual requirements depend on the system's deployment environment, the number of users, the amount of stored data, and the configuration of the server and database.

4.5.1 Hardware Requirements

The hardware requirements identify the minimum and recommended computer specifications for developing and accessing the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. These specifications provide a reference for selecting equipment capable of running the development tools, web application, and database software. The actual hardware requirements may vary depending on the deployment setup, number of concurrent users, and volume of stored records.

Table 4.1. Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (8th Gen) / AMD Ryzen 3	Intel Core i3 (8th Gen) / AMD Ryzen 3
Memory (RAM)	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Graphics	Integrated Graphics	Integrated or Dedicated Graphics Card
Display	1366 × 768 Resolution	1920 × 1080 Full HD
Network	Stable Internet Connection	High-Speed Broadband Internet
Input Devices	Keyboard and Mouse	Keyboard and Mouse
Operating Device	Desktop or Laptop Computer	Desktop or Laptop Computer

The minimum specifications provide a general reference for running the application and accessing its functions through a compatible web browser. The recommended specifications provide additional resources for development activities and for running multiple applications simultaneously.

An SSD is recommended because it generally provides faster file access and application loading than a traditional hard disk drive. Sufficient RAM and processing capability are also useful when running development tools, the Node.js server, and the MySQL database on the same computer.

For ordinary users who only access the deployed application through a web browser, the required hardware may be lower than the specifications needed for development. If the application and database are hosted on a separate server, the server's hardware requirements should be assessed separately according to its workload and expected number of users.

4.5.2 Software Requirements

The development of the Web-Based Lot Availability Monitoring System utilizes modern web technologies that support responsive user interfaces, efficient backend processing, secure database management, and reliable email notification services.

Software	Purpose
Windows 10 / Windows 11	Operating System used for development and testing
Visual Studio Code	Source code editor for frontend and backend development
React.js	Frontend JavaScript library for building the user interface
Tailwind CSS	CSS framework for responsive and modern interface design
Node.js	JavaScript runtime environment for server-side processing
Express.js	Backend framework for creating RESTful APIs
MySQL	Relational Database Management System
MySQL Workbench	Database administration and management
XAMPP *(optional if used)*	Local server environment for MySQL services
Git	Version control system for source code management
GitHub *(if used)*	Online repository for project collaboration and backup
Postman	API testing and debugging tool
Nodemailer SMTP	Email notification service integration
Google Chrome / Microsoft Edge	Recommended web browsers for system access

The selected software tools work together to provide a complete development environment for the proposed application. React and Tailwind CSS are responsible for the client-side interface, while Node.js and Express.js handle business logic and communication with the database. MySQL serves as the centralized repository for all system data, and Nodemailer SMTP enables automated email notifications. Development and testing activities are performed using Visual Studio Code, Postman, and modern web browsers to ensure that the application functions correctly before deployment.

4.6 Database Design

The database serves as the central repository of the Web-Based Lot Availability Monitoring System. It stores all information required for managing users, properties, residential lots, customers, and transactions. The system utilizes MySQL, a relational database management system that organizes data into related tables to ensure consistency, integrity, and efficient retrieval of information.

The database is designed following the principles of relational database management by establishing relationships through primary keys and foreign keys. This structure minimizes data redundancy, maintains referential integrity, and supports accurate processing of system operations. Each table performs a specific function while remaining interconnected with other tables within the database.

Users Table

The Users Table stores account credentials and user information required for system authentication and access control. Each user is assigned a role that determines the functions and information accessible within the system.

Field	Description
user_id	Primary Key; uniquely identifies each user account
fullname	Complete name of the user
username	Login username used to access the system
password	Encrypted password used for authentication
role	User access level or assigned system role
email	Email address associated with the user account
status	Indicates whether the user account is active or inactive
created_at	Date and time when the user account is created

Properties Table

The Properties Table stores information about the real estate properties or subdivision projects managed by Golden Dragon Estate Corporation. It provides the primary property information associated with the residential lots monitored by the system.

Field	Description
property_id	Primary Key; uniquely identifies each property
property_name	Name of the property or subdivision
location	Location or address of the property
description	General description of the property
total_lots	Total number of residential lots within the property
created_at	Date and time when the property record is created
updated_at	Date and time when the property record is last updated

Lots Table

The Lots Table stores detailed information about the individual residential lots managed by the company. This table serves as one of the primary components of the system because it contains the current availability status of each lot.

Field	Description
lot_id	Primary Key; uniquely identifies each lot
property_id	Foreign Key; identifies the property where the lot belongs
block_number	Identifies the block where the lot is located
lot_number	Identifies the specific lot within the block
lot_area	Total area of the residential lot
price	Selling price of the lot
status	Current lot status: Available, Pending, or Sold
created_at	Date and time when the lot record is created
updated_at	Date and time when the lot information is last updated

The property_id establishes the relationship between the Properties Table and the Lots Table. This allows multiple lots to be associated with a specific property or subdivision.

Customers Table

The Customers Table stores customer information required for inquiries, reservations, and lot-related transactions. Centralizing customer information allows authorized personnel to retrieve and maintain customer records efficiently.

Field	Description
customer_id	Primary Key; uniquely identifies each customer
fullname	Complete name of the customer
contact_number	Customer's contact or mobile number
email	Customer's email address
address	Customer's residential or mailing address
created_at	Date and time when the customer record is created
updated_at	Date and time when the customer information is last updated

Transactions Table

The Transactions Table records lot-related transactions performed within the system. It establishes the relationship between customers and the residential lots involved in reservations or sales. The table also provides a historical record that can support reporting and transaction monitoring.

Field	**Description**
transaction_id	Primary Key; uniquely identifies each transaction
customer_id	Foreign Key; identifies the customer involved in the transaction
lot_id	Foreign Key; identifies the residential lot involved
user_id	Foreign Key; identifies the user who records or manages the transaction

transaction_date	Date when the transaction is recorded
transaction_type	Type of transaction, such as reservation or sale
status	Current status of the transaction
amount	Transaction amount or recorded selling amount
remarks	Additional information regarding the transaction
created_at	Date and time when the transaction record is created

The customer_id, lot_id and use_id fields establish relationships with the Customers, Lots, and Users tables. These relationships allow the system to identify which customer is associated with a specific lot transaction and which authorized user records or manages the transaction.

Database Relationship Summary

The tables within the database are interconnected to ensure that information is properly associated across the system. The Properties Table has a one-to-many relationship with the Lots Table, because one property can contain multiple residential lots. The Customers Table is connected to the Transactions Table, allowing the system to identify the customer associated with each transaction. Similarly, the Lots Table is connected to the Transactions Table, allowing the system to maintain a record of transactions associated with individual lots.

The Users Table is also connected to the Transactions Table, allowing the system to identify the authorized personnel responsible for recording or managing transactions. Through these relationships, the database maintains organized and traceable records while reducing unnecessary duplication of information.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of findings, conclusions, recommendations, and impact of the study entitled *Design and Development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation*. The study addressed the limitations of the company's manual process of monitoring residential lot availability through printed subdivision maps and handwritten status updates.

The developed system provides a centralized web-based platform for managing residential lot information and related records. Its implemented functionalities include lot availability monitoring, user authentication, role-based access control, and other property-related management functions supported by the application. These features were developed to help authorized personnel access and maintain lot information through a digital interface.

This chapter summarizes the major findings based on the system's development and actual evaluation results, presents conclusions in relation to the objectives of the study, and recommends improvements for future versions. It also discusses the potential organizational, employee, technological, and academic contributions of the project.

5.2 Summary of Findings

The study focused on developing a web-based system to address the challenges associated with the manual monitoring of residential lot availability at Golden Dragon Estate Corporation. The findings below summarize the system's implemented functions and their intended contributions to the company's operations. Claims about measured improvements should be supported by the actual testing and evaluation results.

5.2.1 Digitalization of Manual Lot Monitoring

The existing process relied on printed subdivision maps and handwritten updates to indicate whether residential lots were available, pending, or sold. Maintaining these physical records required personnel to update lot statuses manually and keep the information organized.

The developed system provides a digital alternative through which authorized users can view and update lot information using a web interface. This approach reduces dependence on manually modified printed maps and provides a centralized location for accessing recorded lot statuses.

5.2.2 Centralized Data Management

The system uses a MySQL relational database to store and organize the information required by its implemented functions. Related records are structured through database tables and relationships, supporting the retrieval and maintenance of information within the application.

Centralizing these records provides a consistent location for accessing stored information and reduces reliance on separate physical documents. The extent to which the system reduces duplicate records or data-entry errors should be determined through actual testing and operational observations.

5.2.3 Interactive Lot Availability Monitoring

The system provides an interface for displaying residential lots and their recorded availability statuses. Color-coded indicators help users distinguish between available, pending, and sold lots. Green represents available lots, yellow represents pending lots, and red represents sold lots.

This visual presentation helps users identify lot statuses without relying solely on handwritten annotations on printed maps. Authorized users can also update lot information through the application according to their assigned permissions.

5.2.4 User Authentication and Access Control

The system incorporates user authentication and role-based access control to regulate access to protected functions. Users must authenticate before accessing the functions that require a valid account, while permissions determine which operations are available to each role.

These mechanisms provide a structured approach to managing user access. Their effectiveness in preventing unauthorized access should be assessed through the actual security checks and test results conducted during system evaluation.

5.2.5 Customer and Transaction Management

The system includes functions for maintaining customer records and recording lot-related transactions, where these modules are implemented in the deployed application. These functions provide an organized way to manage related information and retrieve records when required.

Associating transaction records with relevant customer and lot information can help personnel review recorded business activities. The accuracy and completeness of these records depend on proper data entry, validation, and the database rules implemented in the system.

5.2.6 Dashboard and Reporting

The dashboard presents selected operational information in a centralized interface, helping authorized users review relevant system records. The reporting functions, where implemented, provide summaries of information maintained by the application.

These features can support routine monitoring and reduce the need to manually compile information from separate records. Their contribution to management decision-making depends on the accuracy, completeness, and relevance of the information displayed.

5.2.7 Email Notification

The system incorporates Nodemailer with SMTP for email notifications concerning selected system events, provided that this functionality is configured and operational in the deployed application. This feature offers an additional communication channel for events covered by the notification process.

The reliability and timeliness of email delivery should be assessed through the notification tests performed during system evaluation.

5.2.8 Overall System Development

The study resulted in the development of a web-based application intended to improve the organization and monitoring of residential lot information at Golden Dragon Estate Corporation. The system uses React and Tailwind CSS for the user interface, Node.js and Express.js for server-side operations, and MySQL for database management. Nodemailer with SMTP supports email notifications where configured.

The developed application provides a digital alternative to the company's previous manual lot monitoring process. The extent to which it meets its functional and non-functional requirements should be established through the documented testing results and evaluation findings.

5.3 Conclusion

The study focused on the design and development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. The developed application provides a centralized platform for viewing and managing residential lot information, offering an alternative to the company's previous reliance on printed subdivision maps and handwritten status updates.

The system's interactive lot monitoring interface allows authorized users to view recorded lot statuses through color-coded indicators and update information through the web application. Its relational database organizes lot information and other supported records, while user authentication and role-based access control provide mechanisms for managing access to system functions.

The system also incorporates related information management functions, such as customer records, transaction recording, dashboard information, reporting, and email notifications, where implemented. These functions provide a foundation for organizing operational records within a single application rather than relying exclusively on separate physical documents.

The study demonstrates how web development technologies can be applied to address an identified organizational need. However, the extent to which the system improves efficiency, usability, reliability, and security must be determined from the actual testing and evaluation results rather than inferred from the presence of these features alone.

Overall, the developed system provides Golden Dragon Estate Corporation with a digital platform for lot availability monitoring and related information management. Its continued effectiveness will depend on appropriate system use, data maintenance, security practices, and future improvements based on organizational needs and evaluation findings.

5.4 Recommendations

Based on the development of the system and the operational needs addressed by the study, the following recommendations are proposed to support continued use and future enhancement of the application.

5.4.1 System Improvements

The following improvements may be considered for future versions of the system:

1. **Enhance the interactive lot map.** Additional search, filtering, zooming, and lot-detail features may help users locate specific residential lots and review their information more conveniently.
2. **Expand reporting and analytics.** Additional reports and visual summaries may be developed to present lot availability, transaction histories, and other relevant operational information.
3. **Implement activity logs.** An audit trail may be added to record important actions, such as lot status changes, customer record updates, and transaction modifications. This can help authorized personnel review changes and improve accountability.
4. **Strengthen database backup and recovery.** A structured backup schedule and tested recovery procedures should be established to reduce the risk of losing important records because of accidental deletion, hardware failure, or other incidents.
5. **Expand email notification options.** If email notifications are part of the deployed system, future versions may provide additional notification settings for supported lot and transaction events.

1. **Improve mobile responsiveness.** Further interface optimization for smartphones and tablets may make it easier for authorized personnel to access lot information using different devices.
2. **Enhance security mechanisms.** Future improvements may include stronger password policies, improved session management, multi-factor authentication, and additional security monitoring, depending on the system's existing safeguards and the organization's needs.

5.4.2 Future Research Directions

Future researchers may explore additional technologies and evaluation methods to extend the capabilities of web-based lot availability monitoring systems.

First, Geographic Information System (GIS) technology may be investigated to support more advanced mapping and geographical visualization of properties and individual lots.

Second, future studies may examine cloud-based hosting and database infrastructure to assess their suitability for improving accessibility, scalability, backup management, and system availability.

Third, a mobile application or an enhanced mobile-friendly interface may be developed to provide convenient access to lot information through smartphones and tablets.

Fourth, researchers may explore predictive analytics using historical lot availability and transaction data. Such research could investigate whether data analysis can help identify trends and provide useful information for business planning. The usefulness of these techniques would need to be established through appropriate analysis and evaluation.

Finally, future studies may conduct longer-term evaluations involving additional users and a wider range of operational scenarios. These evaluations could examine usability, reliability, efficiency, security, and user satisfaction using clearly defined evaluation instruments and documented results.

5.4.3 Implementation Strategies

To support the effective use and maintenance of the system, Golden Dragon Estate Corporation should consider the following implementation strategies.

First, employees should receive appropriate training based on their responsibilities and the functions available to their assigned roles. Training may cover authentication, lot monitoring, status updating, and other relevant record management functions.

Second, user accounts and permissions should be reviewed regularly to ensure that access corresponds to each employee's current responsibilities. Accounts that are no longer required should be handled according to the organization's access management procedures.

Third, the organization should establish database backup and recovery procedures. Designated personnel should verify that backups are completed successfully and periodically check whether the data can be restored when necessary.

Fourth, routine maintenance should be planned to address software defects, dependency updates, database issues, and security concerns. Changes should be reviewed and tested before being applied to the operational system to minimize disruption.

Finally, feedback should be gathered from employees who use the application during actual operations. Their observations can help identify usability concerns, missing functions, and opportunities for improvement. The organization can use this feedback to prioritize future enhancements according to operational needs and available resources.

5.5 Impact of the Study

The study contributes a digital approach to managing residential lot availability and related information at Golden Dragon Estate Corporation. Its potential impact can be considered in terms of organizational operations, employee activities, data management, technology, and academic research.

5.5.1 Organizational Impact

The system provides a centralized platform for accessing and maintaining lot information. By offering a digital alternative to printed subdivision maps and handwritten status updates, it supports the organization's efforts to organize its lot monitoring activities.

The availability of organized lot information may also assist personnel in reviewing the current recorded status of residential lots. The actual extent of improvement in operational efficiency should be determined through evaluation results and observations of system use.

5.5.2 Employee Impact

The application gives authorized employees a digital interface for accessing lot information and performing supported operational tasks. Color-coded indicators provide a visual means of distinguishing recorded lot statuses, while centralized records can make information easier to retrieve.

Appropriate training and clearly defined access permissions remain important to ensure that employees can use the system correctly and access only the functions required for their responsibilities.

5.5.3 Data Management Impact

The use of a relational database provides a structured way to store and retrieve lot information and other supported records. Relationships among database tables help organize related information and provide a foundation for consistent record management.

The benefits of this structure depend on proper validation, accurate data entry, appropriate access controls, and regular maintenance of the database.

5.5.4 Technological Impact

The project demonstrates the application of modern web development technologies to an operational problem in real estate management. React, Tailwind CSS, Node.js, Express.js, and MySQL provide the technological foundation for the user interface, server-side processing, and data storage. Nodemailer with SMTP supports email notifications where implemented.

The system's organization into distinct components also provides a basis for future development, provided that enhancements are designed and tested for compatibility with the existing application.

5.5.5 Academic Impact

The study provides a reference for students and future researchers interested in web-based property monitoring, inventory management, and information systems development. Its system requirements, architecture, database design, and development approach may inform similar projects addressing the management of organizational records.

Future researchers may build upon this work by conducting additional evaluations, examining alternative system architectures, or investigating technologies that support more advanced property monitoring and reporting.

5.6 Summary

This study involved the development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation to address the limitations of its manual lot monitoring process. The application provides a digital platform for viewing and updating residential lot statuses and managing related information through the functions implemented in the system.

The system uses a web-based interface, server-side application logic, and a relational database to organize lot information and other supported records. Authentication and role-based access control provide mechanisms for managing user access, while additional functions support operational record management and communication where implemented.

The study demonstrates the practical application of web technologies to the organization's lot monitoring needs. Recommendations for future development include enhancing the interactive lot map, expanding reporting,

implementing activity logs, strengthening backup and recovery procedures, improving mobile usability, and reviewing security mechanisms.

Future research may examine GIS integration, cloud infrastructure, mobile access, predictive analytics, and longer-term system evaluation. These opportunities can help guide the continued development of the application according to the organization's requirements.

Overall, the developed system provides a centralized digital alternative for residential lot availability monitoring. Its long-term value will depend on accurate record maintenance, appropriate use, effective security practices, and continued assessment of its performance and suitability for the organization.